



INN HOTELS Group

Business Report

Machine Learning Based Solution To Predict Booking Cancellations

PGPDSGA

Submitted By

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I. Problem Context:

A significant number of hotel bookings are called off due to cancellations or no-shows. The typical reasons for cancellations include change of plans, scheduling conflicts, etc. This is often made easier by the option to do so free of charge or preferably at a low cost, which is beneficial to hotel guests, but it is a less desirable and possibly revenue-diminishing factor for hotels to deal with. Such losses are particularly high on last-minute cancellations.

The new technologies involving online booking channels have dramatically changed customers' booking possibilities and behaviour. This adds a further dimension to the challenge of how hotels handle cancellations, which are no longer limited to traditional booking and guest characteristics.

The cancellation of bookings impacts a hotel on various fronts:

1. Loss of resources (revenue) when the hotel cannot resell the room.
2. Additional costs of distribution channels by increasing commissions or paying for publicity to help sell these rooms.
3. Lowering prices last minute, so the hotel can resell a room, resulting in a reduction of the profit margin.
4. Human resources to make arrangements for the guests.

II. Objective:

The increasing number of cancellations calls for a Machine Learning based solution that can help in predicting which booking is likely to be cancelled. INN Hotels Group has a chain of hotels in Portugal, and they are facing problems with the high number of booking cancellations and have reached out to your firm for data-driven solutions. As a data scientist have to analyze the data provided to find which factors have a high influence on booking cancellations, build a predictive model that can predict which bookings are going to be cancelled in advance, and help in formulating profitable policies for cancellations and refunds.

III. Problem Definition:

Minimize loss by booking cancellations

1. Able to predict Booking Cancellation.
2. Find factors influencing booking cancellation
3. Able to do resell rooms by various methods

Find answers to **EDA Questions**:

1. What are the busiest months in the hotel?
2. Which market segment do most of the guests come from?
3. Hotel rates are dynamic and change according to demand and customer demographics. What are the differences in room prices in different market segments?
4. What percentage of bookings are cancelled?
5. Repeating guests are the guests who stay in the hotel often and are important to brand equity. What percentage of repeating guests cancel?
6. Many guests have special requirements when booking a hotel room. Do these requirements affect booking cancellation?

IV. Exploratory Data Analysis

IV.1. Data Description:

The data contains the different attributes of customers' booking details. The detailed data dictionary is given below.

Data Dictionary:

- Booking_ID: the unique identifier of each booking
- no_of_adults: Number of adults
- no_of_children: Number of Children
- no_of_weekend_nights: Number of weekend nights (Saturday or Sunday) the guest stayed or booked to stay at the hotel
- no_of_week_nights: Number of weeknights (Monday to Friday) the guest stayed or booked to stay at the hotel
- type_of_meal_plan: Type of meal plan booked by the customer:
 - Not Selected – No meal plan selected
 - Meal Plan 1 – Breakfast
 - Meal Plan 2 – Half board (breakfast and one other meal)
 - Meal Plan 3 – Full board (breakfast, lunch, and dinner)

- `required_car_parking_space`: Does the customer require a car parking space? (0 - No, 1- Yes)
- `room_type_reserved`: Type of room reserved by the customer. The values are ciphered (encoded) by INN Hotels Group
- `lead_time`: Number of days between the date of booking and the arrival date
- `arrival_year`: Year of arrival date
- `arrival_month`: Month of arrival date
- `arrival_date`: Date of the month
- `market_segment_type`: Market segment designation.
- `repeated_guest`: Is the customer a repeated guest? (0 - No, 1- Yes)
- `no_of_previous_cancellations`: Number of previous bookings that were canceled by the customer prior to the current booking
- `no_of_previous_bookings_not_canceled`: Number of previous bookings not canceled by the customer prior to the current booking
- `avg_price_per_room`: Average price per day of the reservation; prices of the rooms are dynamic. (in euros)
- `no_of_special_requests`: Total number of special requests made by the customer (e.g. high floor, view from the room, etc)
- `booking_status`: Flag indicating if the booking was canceled or not.

IV.2. Data Quality Checks:

IV.2.1. Missing Values:

There are no missing values in data

IV.2.2. Duplicate Values:

There are no duplicate values in data

IV.3. Univariate Analysis:

IV.3.1. Univariate Analysis For Numerical Columns

IV.3.1.1. Univariate Analysis For No of Adults:

IV.3.1.1.1. Descriptive statistics for no_of_adults:

count	36275.000000
mean	1.844962
std	0.518715
min	0.000000
25%	2.000000
50%	2.000000
75%	2.000000
max	4.000000

Name: no_of_adults, dtype: float64

IV.3.1.1.2. Visual Representation for no_of_adults:

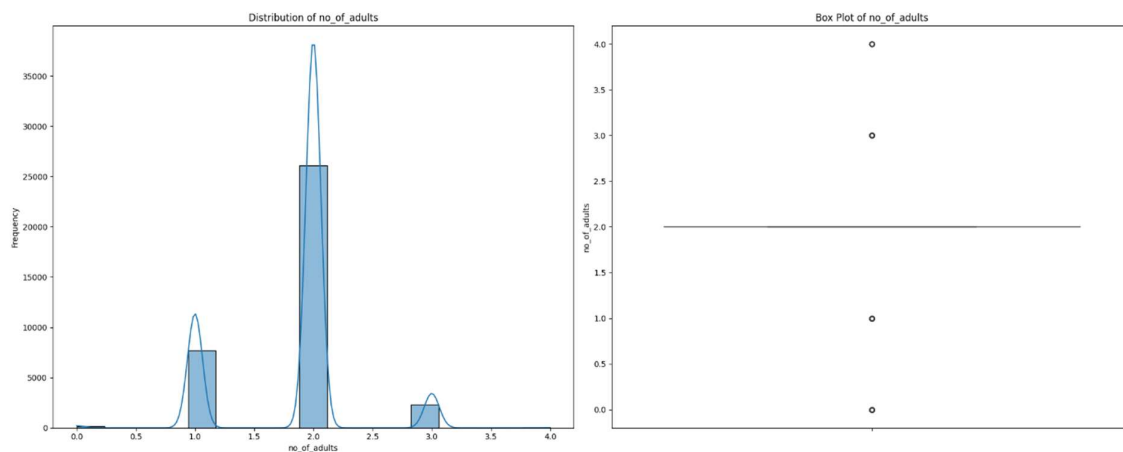
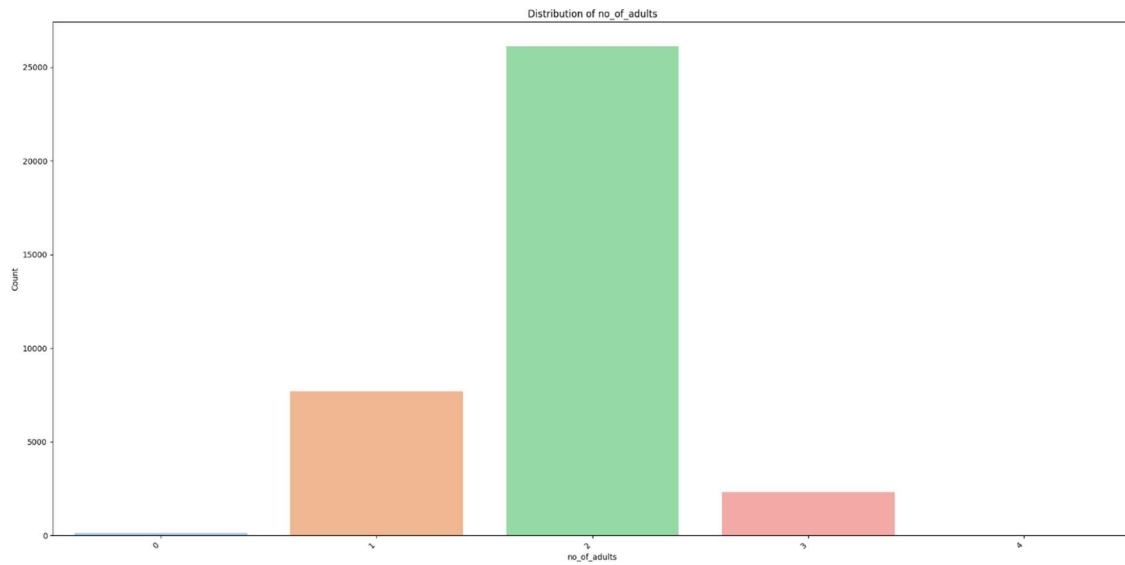


Figure 1 Univariate - No of Adults



IV.3.1.2. Univariate Analysis For No of children

IV.3.1.2.1. Descriptive statistics for no_of_children:

count	36275.000000
mean	0.105279
std	0.402648
min	0.000000
25%	0.000000
50%	0.000000
75%	0.000000
max	10.000000

Name: no_of_children, dtype: float64

IV.3.1.2.2. Visual Representation for no_of_children:

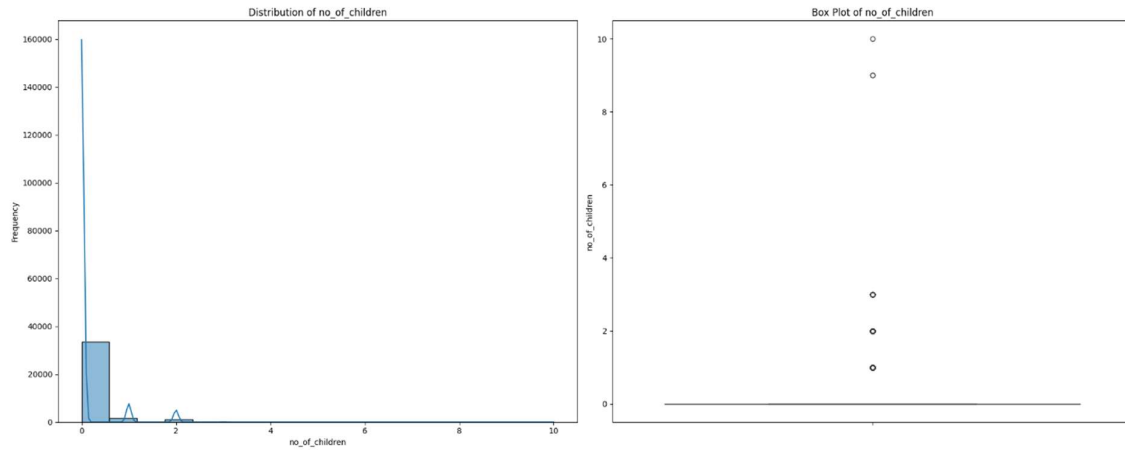
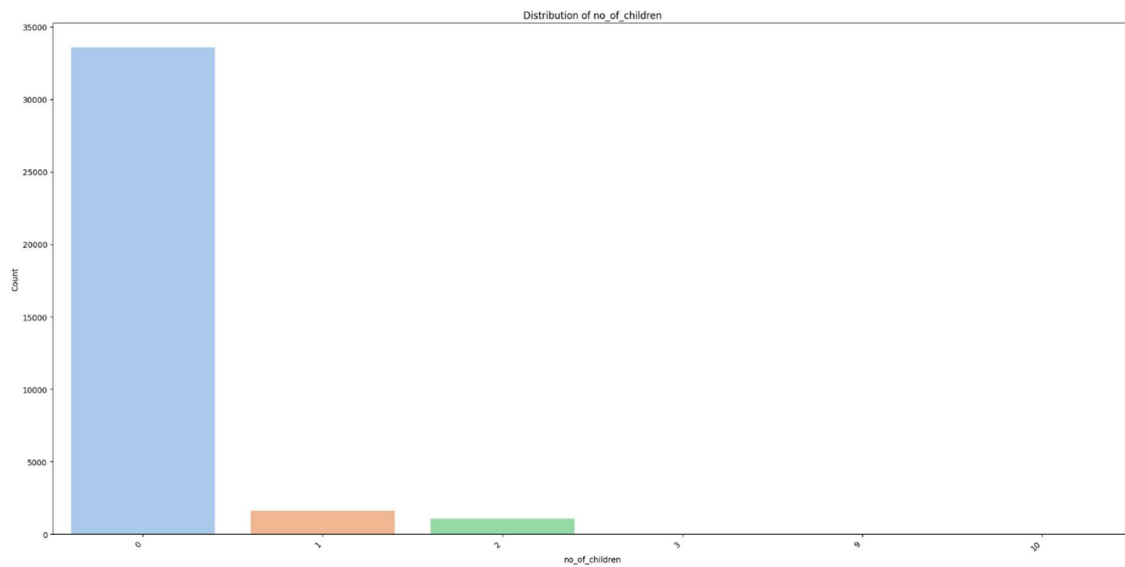


Figure 2 Univariate No of Children



IV.3.1.3. Univariate Analysis For No of weekend nights

IV.3.1.3.1. Descriptive statistics for no_of_weekend_nights:

count	36275.000000
mean	0.810724
std	0.870644
min	0.000000
25%	0.000000
50%	1.000000
75%	2.000000
max	7.000000

Name: no_of_weekend_nights, dtype: float64

IV.3.1.3.2. Visual Representation for no_of_weekend_nights:

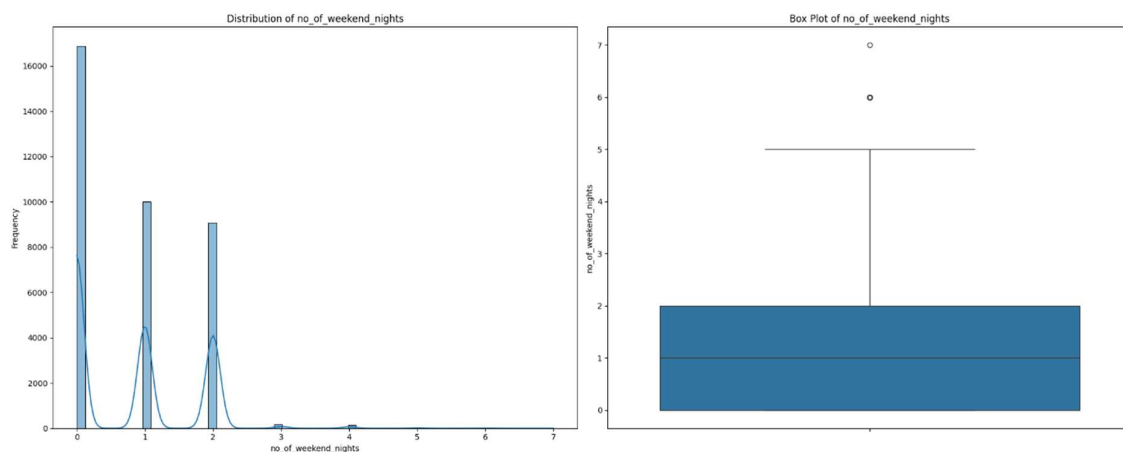
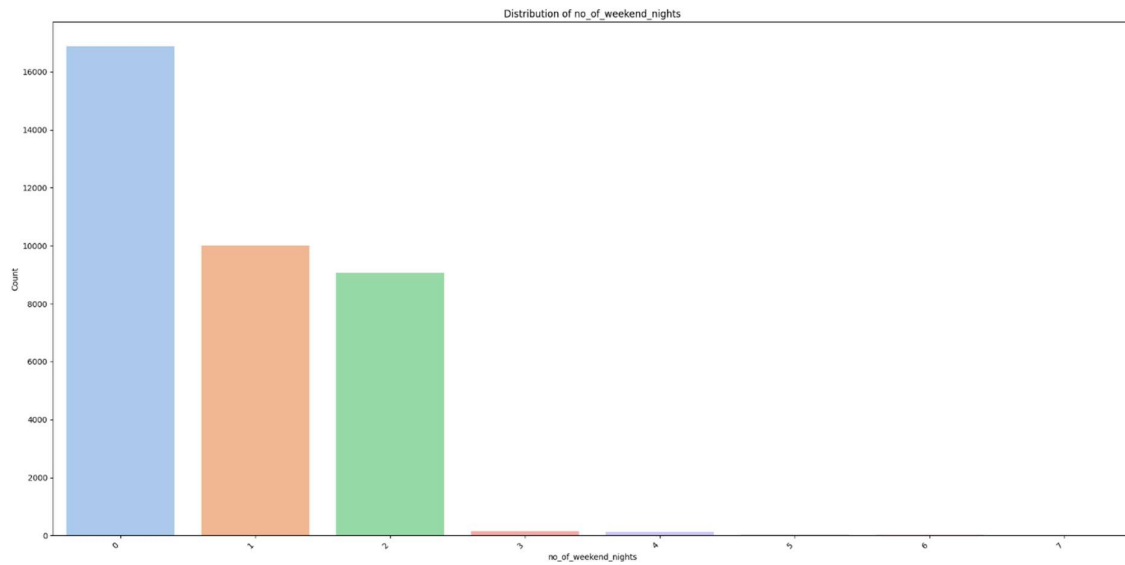


Figure 3 Univariate No of Weekend Nights



IV.3.1.4. Univariate Analysis For No of Week nights

IV.3.1.4.1. Descriptive statistics for no_of_week_nights:

count	36275.000000
mean	2.204300
std	1.410905
min	0.000000
25%	1.000000
50%	2.000000
75%	3.000000
max	17.000000

Name: no_of_week_nights, dtype: float64

IV.3.1.4.2. Visual Representation for no_of_week_nights:

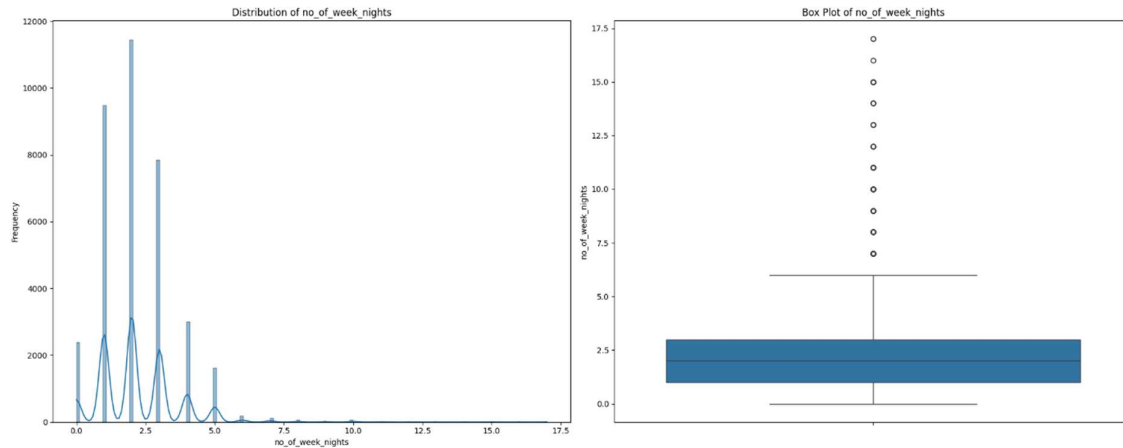
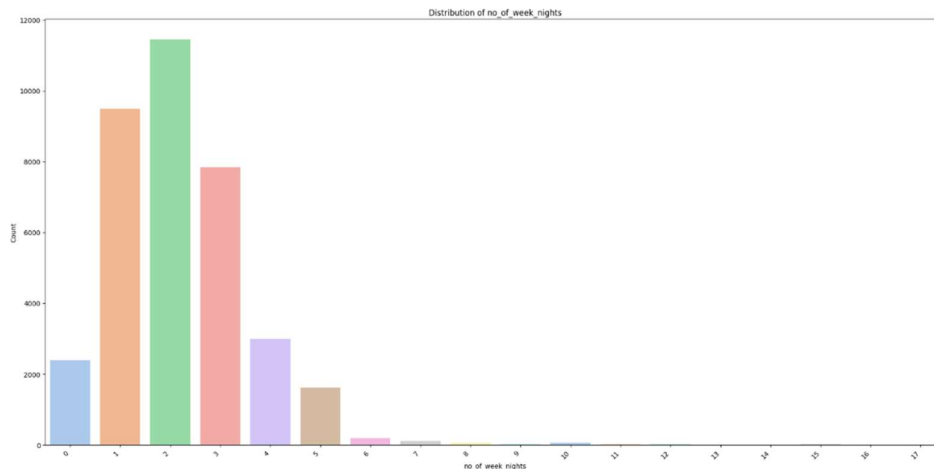


Figure 4 Univariate No of Week Nights



IV.3.1.5. Univariate Analysis For Required car parking Space

IV.3.1.5.1. Descriptive statistics for required_car_parking_space:

count	36275.000000
-------	--------------

mean	0.030986
std	0.173281
min	0.000000
25%	0.000000
50%	0.000000
75%	0.000000
max	1.000000

Name: required_car_parking_space, dtype: float64

IV.3.1.5.2. Visual Representation for required_car_parking_space:

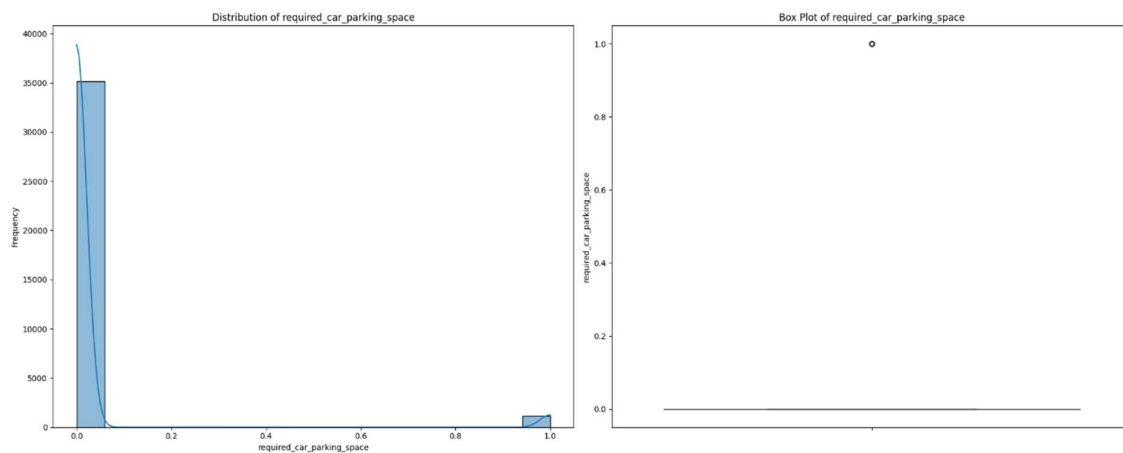
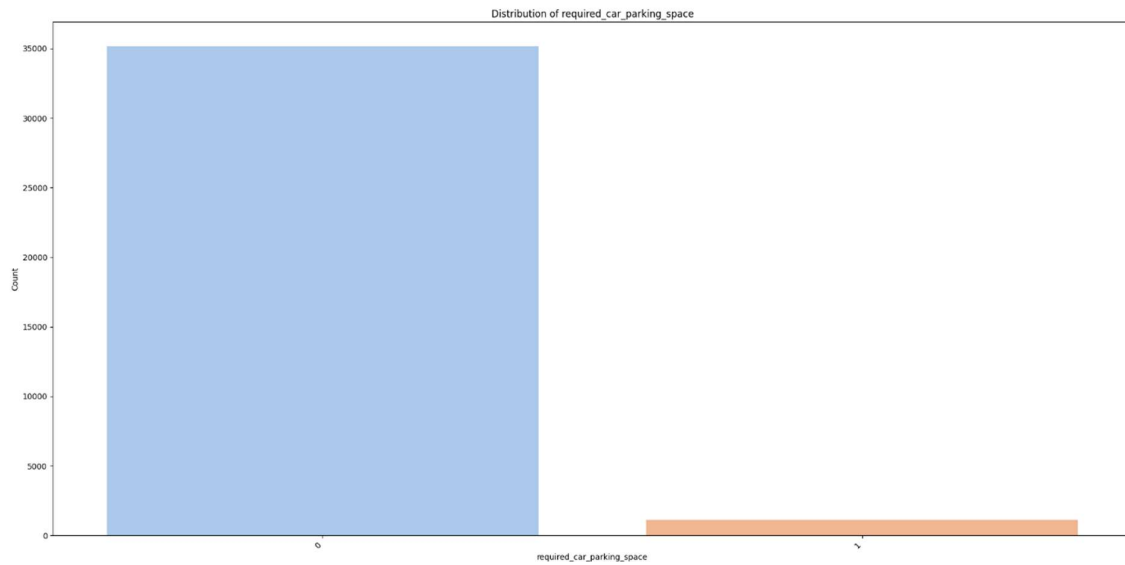


Figure 5 Univariate Required Car Parking Space



IV.3.1.6. Univariate Analysis For Lead Time

IV.3.1.6.1. Descriptive statistics for lead_time:

count	36275.000000
mean	85.232557
std	85.930817
min	0.000000
25%	17.000000
50%	57.000000
75%	126.000000
max	443.000000

Name: lead_time, dtype: float64

IV.3.1.6.2. Visual Representation for lead_time:

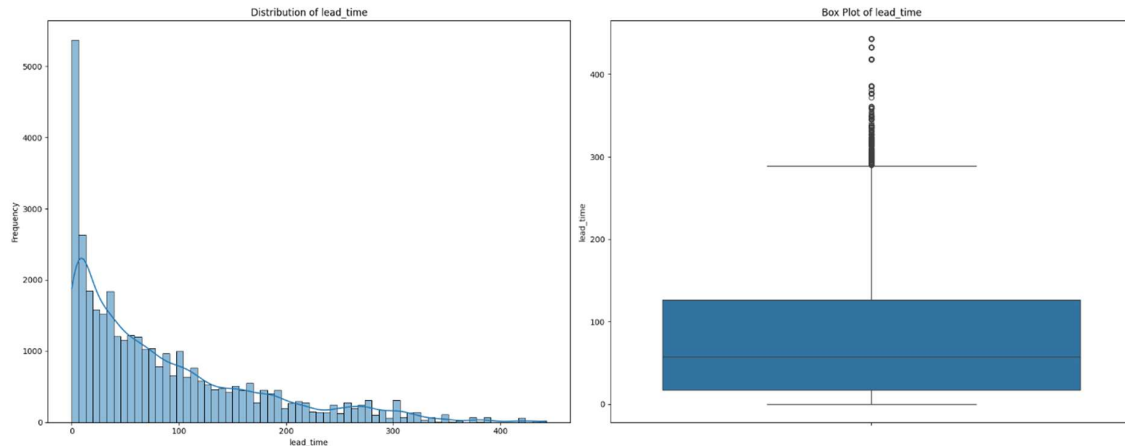
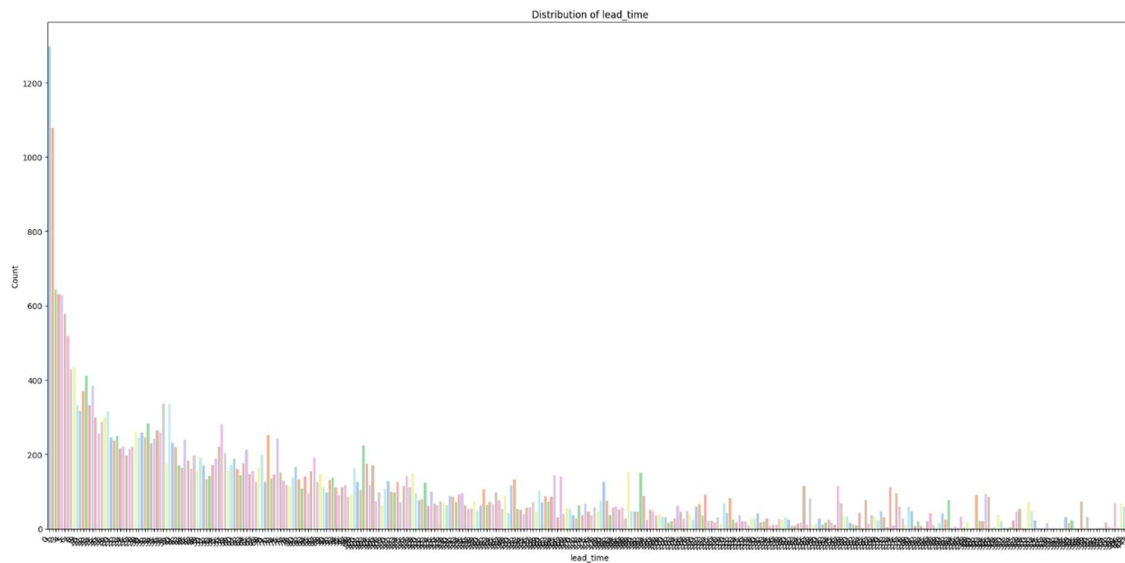


Figure 6 Univariate - Lead Time



IV.3.1.7. Univariate Analysis For Arrival Year

IV.3.1.7.1. Descriptive statistics for arrival_year:

count	36275.000000
-------	--------------

mean	2017.820427
std	0.383836
min	2017.000000
25%	2018.000000
50%	2018.000000
75%	2018.000000
max	2018.000000

Name: arrival_year, dtype: float64

IV.3.1.7.2. Visual Representation for arrival_year:

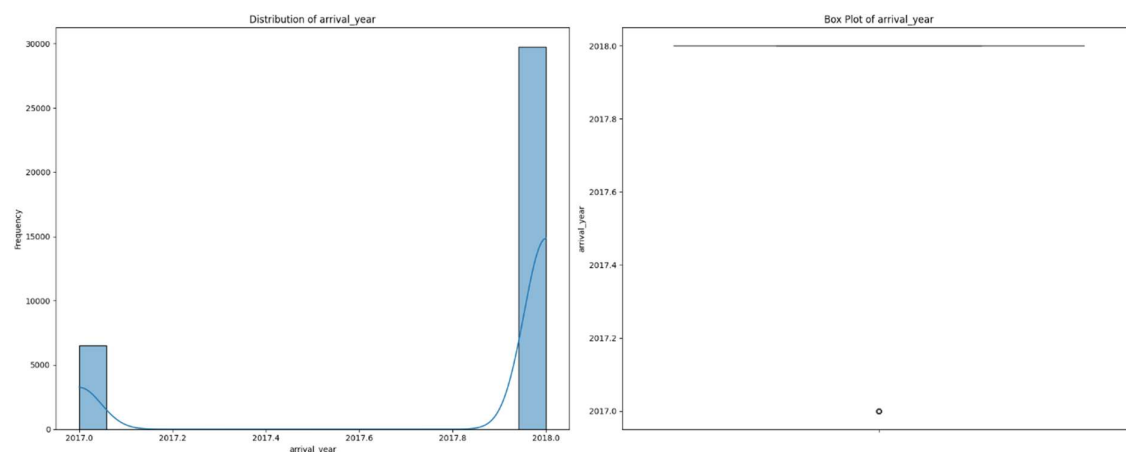
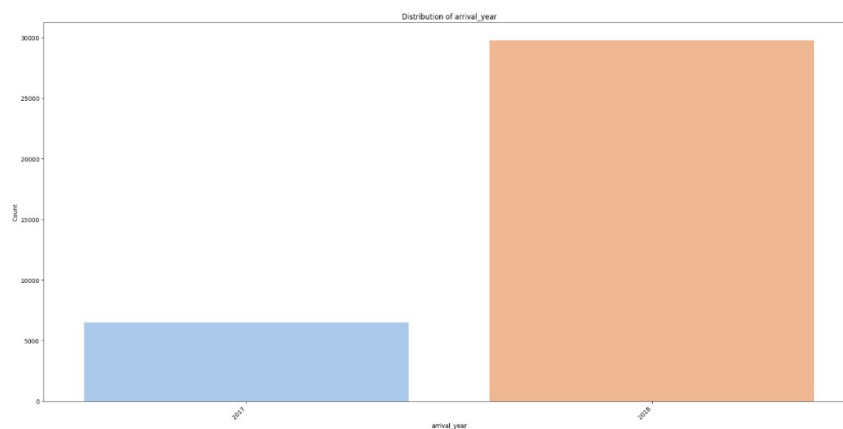


Figure 7 Univariate - Arrival Year



IV.3.1.8. Univariate Analysis For Arrival Month

IV.3.1.8.1. Descriptive statistics for arrival_month:

count	36275.000000
mean	7.423653
std	3.069894
min	1.000000
25%	5.000000
50%	8.000000
75%	10.000000
max	12.000000

Name: arrival_month, dtype: float64

IV.3.1.8.2. Visual Representation for arrival_month:

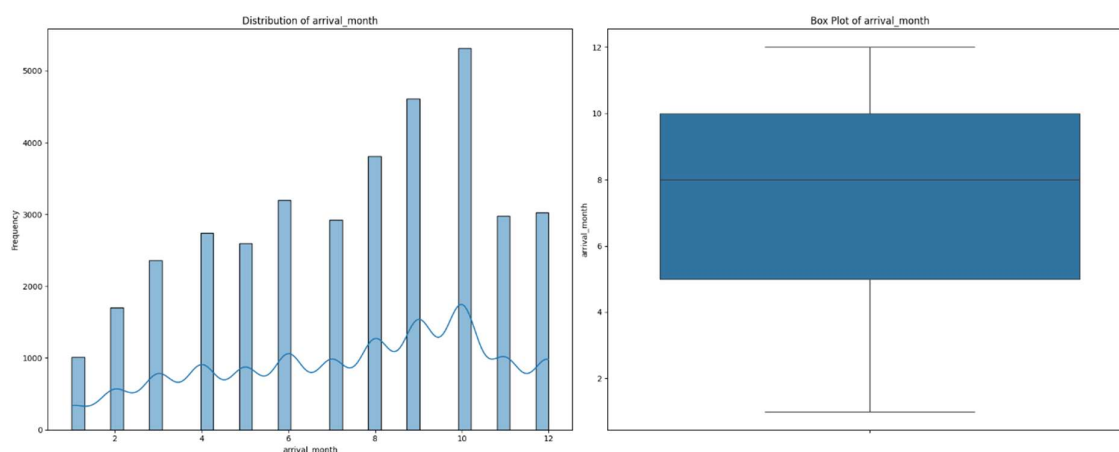
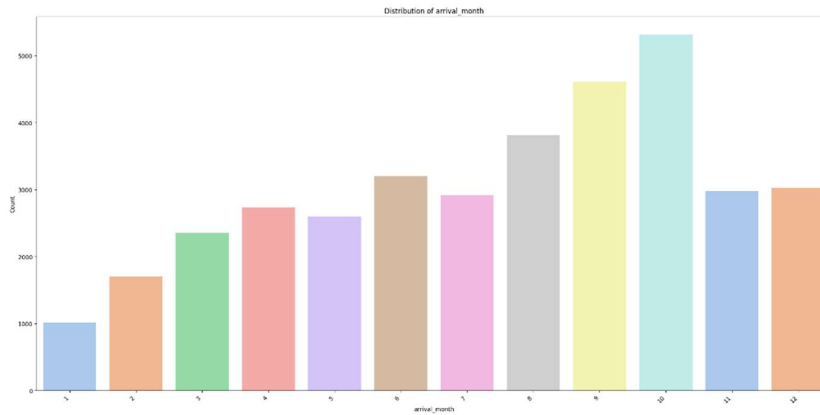


Figure 8 Univariate Arrival Month



IV.3.1.9. Univariate Analysis For Arrival Date

IV.3.1.9.1. Descriptive statistics for arrival_date:

count	36275.000000
mean	15.596995
std	8.740447
min	1.000000
25%	8.000000
50%	16.000000
75%	23.000000
max	31.000000

Name: arrival_date, dtype: float64

IV.3.1.9.2. Visual Representation for arrival_date:

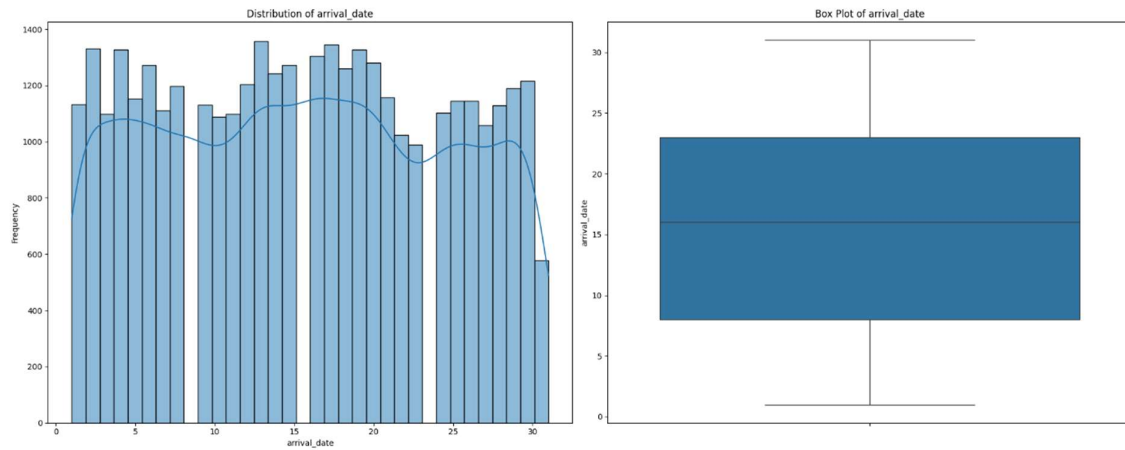
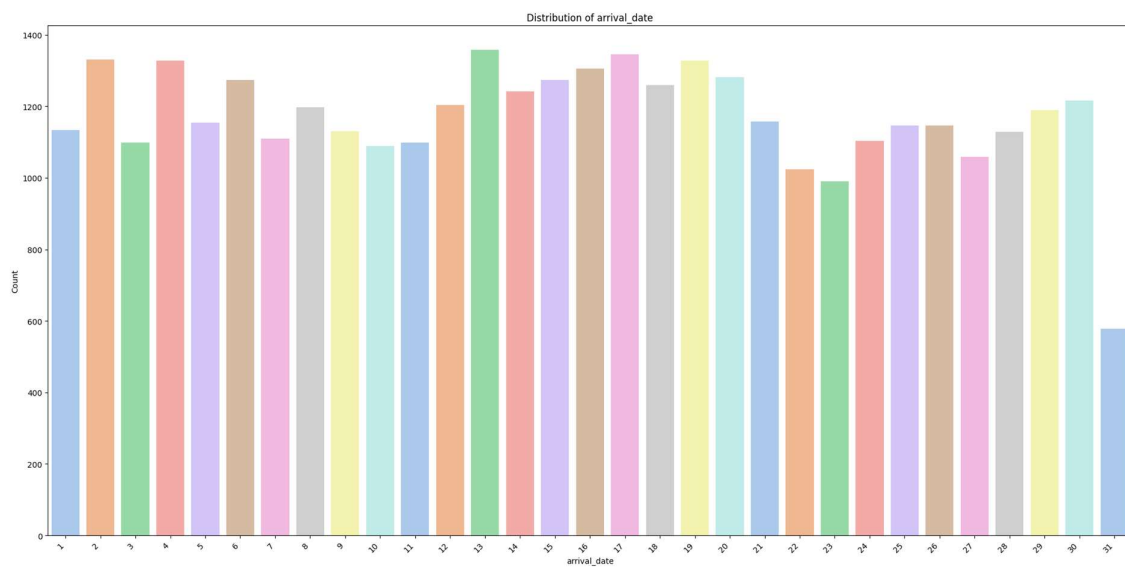


Figure 9 Univariate Arrival Date



IV.3.1.10. Univariate Analysis For Repeated Guests

IV.3.1.10.1. Descriptive statistics for repeated_guest:

count	36275.000000
mean	0.025637
std	0.158053
min	0.000000

25%	0.000000
50%	0.000000
75%	0.000000
max	1.000000

Name: repeated_guest, dtype: float64

IV.3.1.10.2. Visual
Representation for repeated
guest:

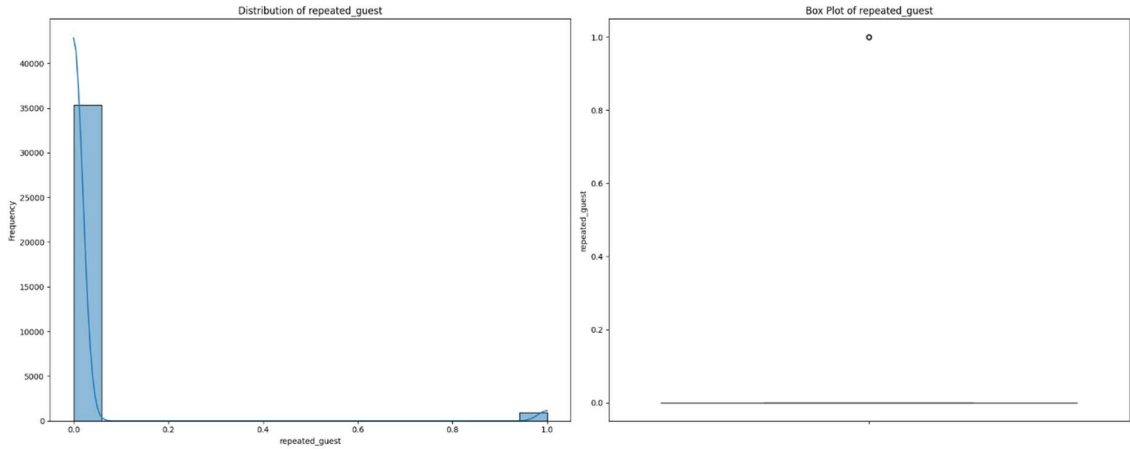
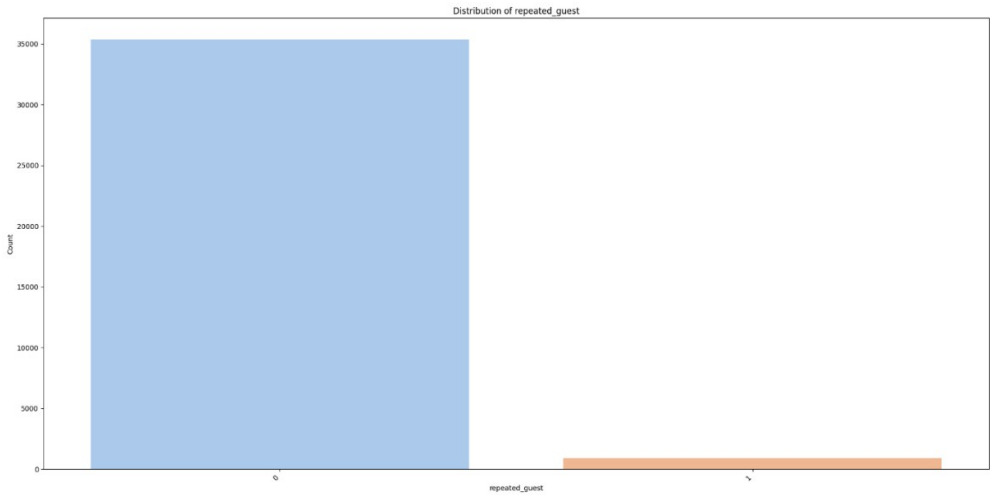


Figure 10 Univariate Repeated Guest



IV.3.1.11. Univariate Analysis For No of Previous Cancellations

IV.3.1.11.1. Descriptive statistics for no_of_previous_cancellations:

count	36275.000000
mean	0.023349
std	0.368331
min	0.000000
25%	0.000000
50%	0.000000
75%	0.000000
max	13.000000

Name: no_of_previous_cancellations, dtype: float64

IV.3.1.11.2. Visual Representation for no_of_previous_cancellations:

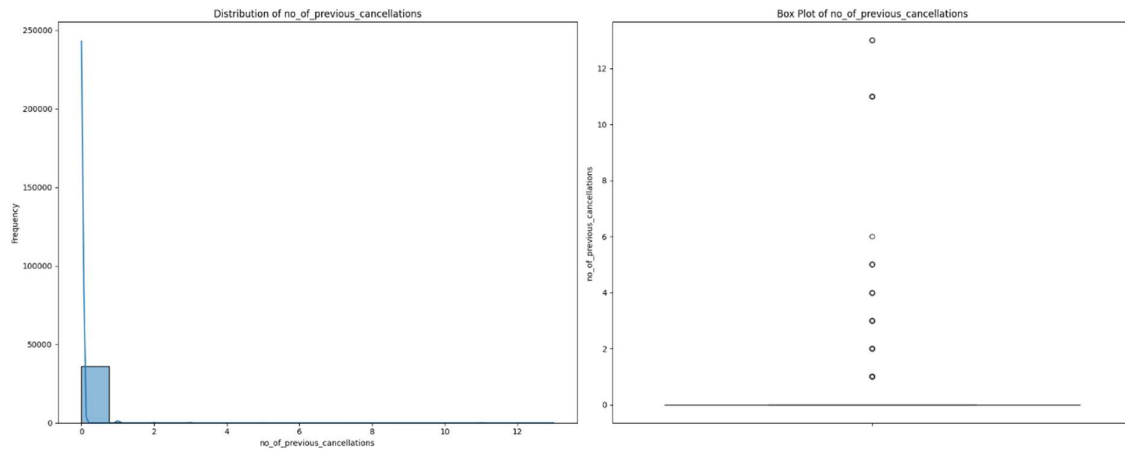
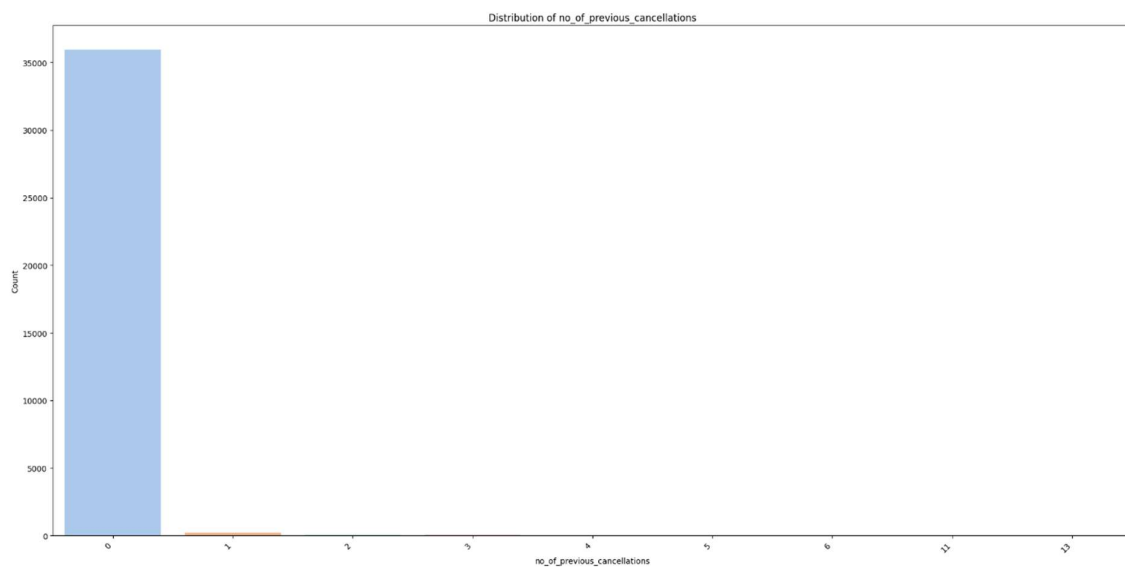


Figure 11 Univariate No of Previous Cancellations



IV.3.1.12. Univariate Analysis For No of Previous Bookings Not Cancelled

IV.3.1.12.1. Descriptive statistics for no_of_previous_bookings_not_canceled:

count	36275.000000
mean	0.153411
std	1.754171

min	0.000000
25%	0.000000
50%	0.000000
75%	0.000000
max	58.000000

Name: no_of_previous_bookings_not_canceled, dtype: float64

IV.3.1.12.2. Visual Representation for
no_of_previous_bookings_not_canceled:

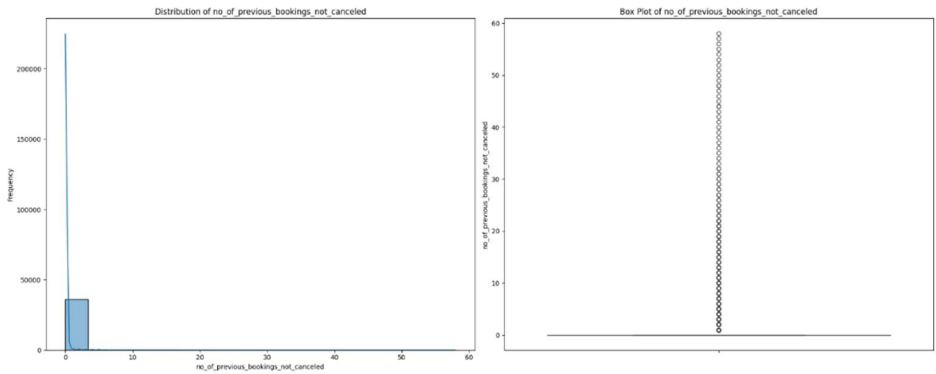
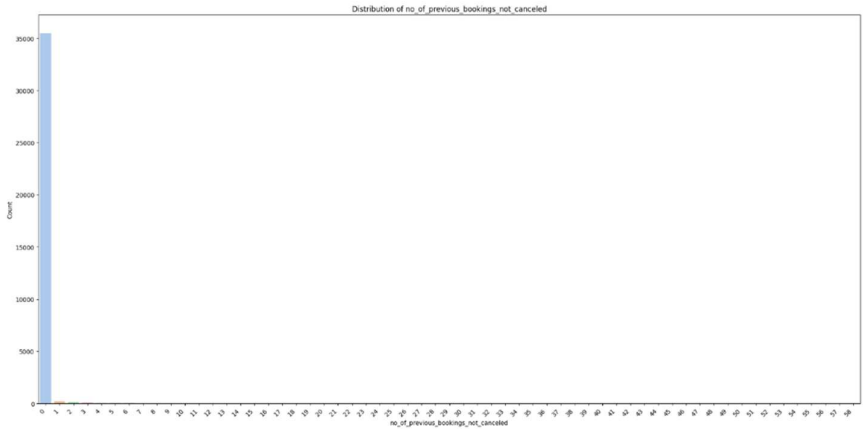


Figure 12 Univariate No of Previous Booking Not Cancelled



IV.3.1.13. Univariate Analysis For Average Room Price

IV.3.1.13.1. Descriptive statistics for avg_price_per_room:

count	36275.000000
mean	103.423539
std	35.089424
min	0.000000
25%	80.300000
50%	99.450000
75%	120.000000
max	540.000000

Name: avg_price_per_room, dtype: float64

IV.3.1.13.2. Visual Representation for avg_price_per_room:

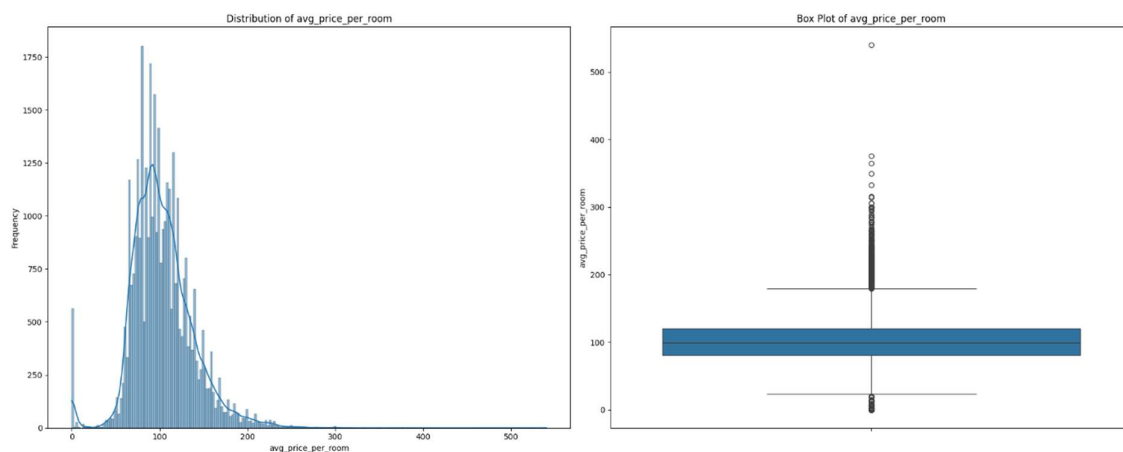
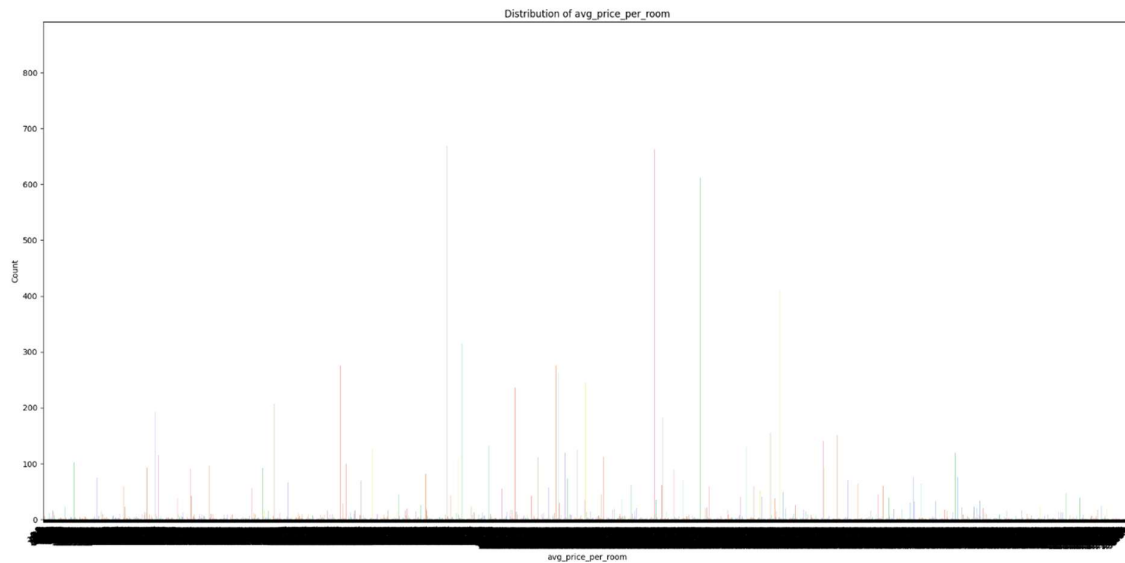


Figure 13 Univariate Avg Price Per Room



IV.3.1.14. Univariate Analysis For No of Special Requests

IV.3.1.14.1. Descriptive statistics for no_of_special_requests:

count	36275.000000
mean	0.619655
std	0.786236
min	0.000000
25%	0.000000
50%	0.000000
75%	1.000000
max	5.000000

Name: no_of_special_requests, dtype: float64

IV.3.1.14.2. Visual Representation for no_of_special_requests:

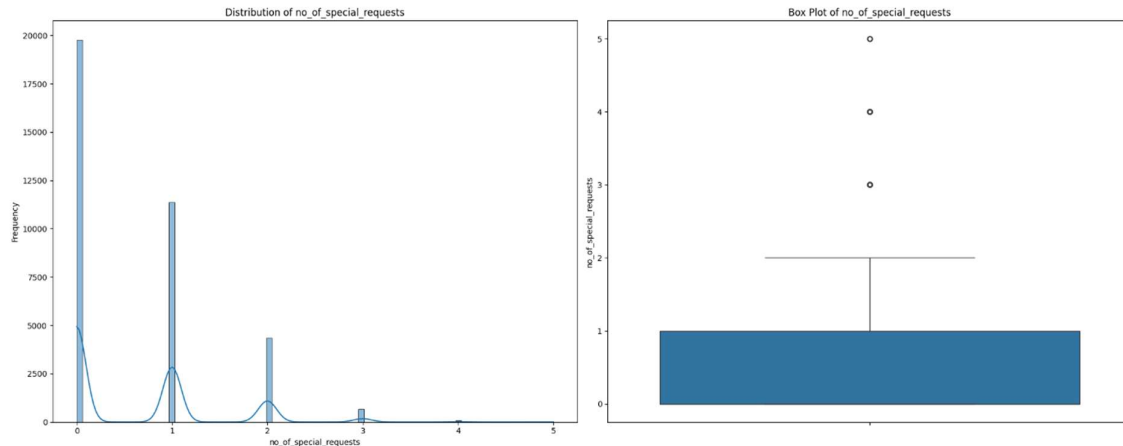
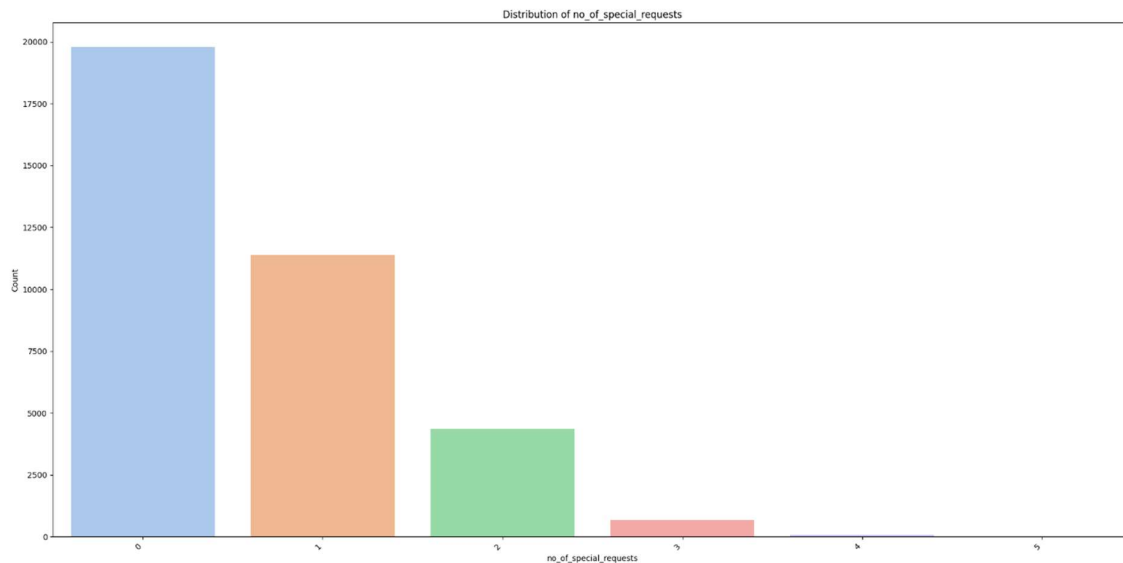


Figure 14 Univariate - No of Special Requests



IV.3.1.15. Inferences from Univariate Analysis for Numerical Columns

no_of_adults: Most bookings are for 2 adults (median, 25th, and 75th percentile are all 2), indicating that couples or small groups are the primary customer segment for INN Hotels. Bookings for more than 2 adults are less frequent, with a maximum of 4 adults per booking.

no_of_children: An overwhelming majority of bookings (75% of bookings have 0 children) do not include children, suggesting that the hotel's clientele is predominantly adults without

accompanying children. While there are some bookings with children, they are a small percentage of the total.

no_of_weekend_nights: Guests tend to book short stays over the weekend, with 75% of bookings including 2 or fewer weekend nights. A significant portion of bookings (25th percentile is 0) involve no weekend nights, indicating a mix of weekday business and leisure travel.

no_of_week_nights: Most stays fall between 1 and 3 weeknights (25th to 75th percentile), pointing to short to medium-length weekday trips. The maximum of 17 weeknights indicates some extended stays, but these are outliers.

required_car_parking_space: Only a very small percentage of guests (around 3%) require a car parking space. This suggests that the majority of guests might be using public transport, ride-sharing, or the hotel's location makes a personal vehicle unnecessary for their stay.

lead_time: Bookings are often made relatively close to the arrival date, with a median lead time of 57 days and 75% of bookings within 126 days. However, there's a wide range, from same-day bookings (0 days) to reservations made over a year in advance (up to 443 days).

arrival_year: All recorded bookings are concentrated within 2017 and 2018, providing a specific two-year window for this dataset. This indicates the scope of the data for analyzing booking trends.

arrival_month: Bookings are distributed throughout the year, with a higher frequency observed in the later months, as indicated by the median arrival month of 8 (August) and the 75th percentile being 10 (October). This suggests potential seasonal peaks during autumn.

arrival_date: Arrival dates are fairly spread out across the month, with the median date being the 16th. There isn't a strong pattern of arrivals concentrated on the beginning or end of the month, indicating a continuous flow of guests.

repeated_guest: The proportion of repeated guests is very low (around 2.5%), highlighting a potential area for INN Hotels to focus on customer loyalty and retention strategies. Most guests are new or infrequent visitors.

no_of_previous_cancellations: Almost all guests (75th percentile is 0) have no prior cancellations, which is a positive sign for booking reliability. The few instances of previous cancellations are outliers and could be associated with specific customer segments.

no_of_previous_bookings_not_canceled: Similar to previous cancellations, the vast majority of guests (75th percentile is 0) have no prior non-canceled bookings, reinforcing the observation of a low repeated guest rate. This further suggests that most bookings come from new customers.

avg_price_per_room: The average price per room is dynamic, with a median of 99.45 euros, ranging from very low prices (0 euros, possibly promotional or special cases) to high values (up to 540 euros). This indicates flexible pricing strategies and diverse customer segments.

no_of_special_requests: Most bookings involve few or no special requests (median is 0, 75th percentile is 1), simplifying operations for the majority of reservations. Some guests do have specific needs, but they are generally limited to one request.

IV.3.2. Univariate Analysis For Categorical columns

IV.3.2.1. Univariate Analysis For Type of Meal Plan

Value Counts for type_of_meal_plan:

Meal Plan 1	27835
Not Selected	5130
Meal Plan 2	3305
Meal Plan 3	5

Name: count, dtype: int64

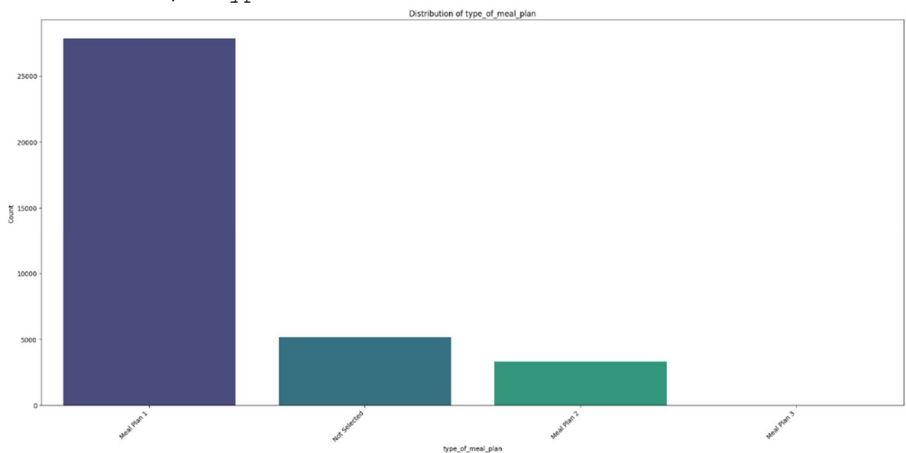


Figure 15 Univariate Type of Meal Plan

IV.3.2.2. Univariate Analysis For Room Type Reserved

Value Counts for room_type_reserved:

room_type_reserved	
Room_Type 1	28130
Room_Type 4	6057
Room_Type 6	966
Room_Type 2	692
Room_Type 5	265
Room_Type 7	158
Room_Type 3	7

Name: count, dtype: int64

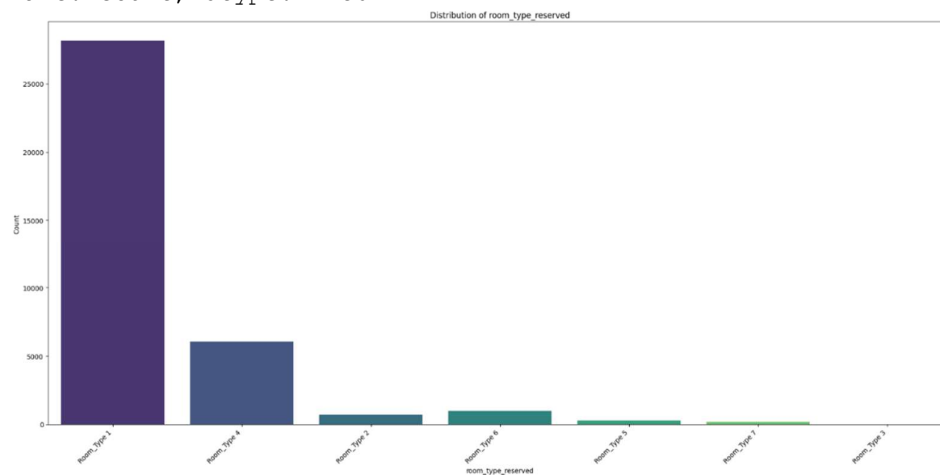


Figure 16 Univariate Room Type Reserved

IV.3.2.3. Univariate Analysis For Market Segment Type

Value Counts for market_segment_type:

market_segment_type	
Online	23214
Offline	10528
Corporate	2017
Complementary	391
Aviation	125

Name: count, dtype: int64

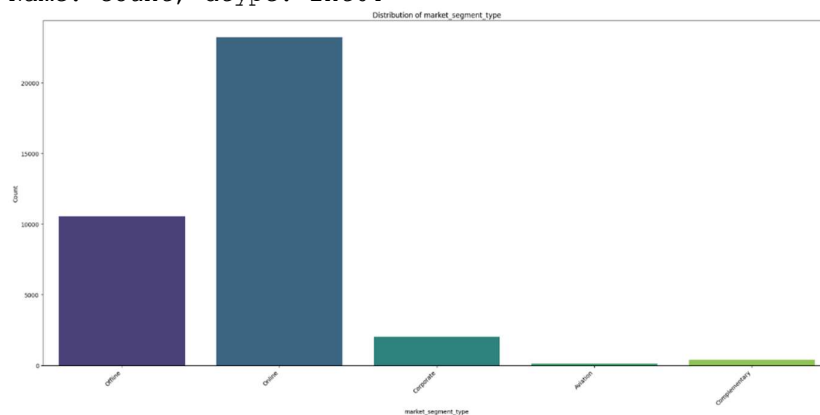


Figure 17 Univariate Market Segment Type

IV.3.2.4. Univariate Analysis For Booking Status

Value Counts for booking_status:

booking_status	
Not_Canceled	24390
Canceled	11885

Name: count, dtype: int64

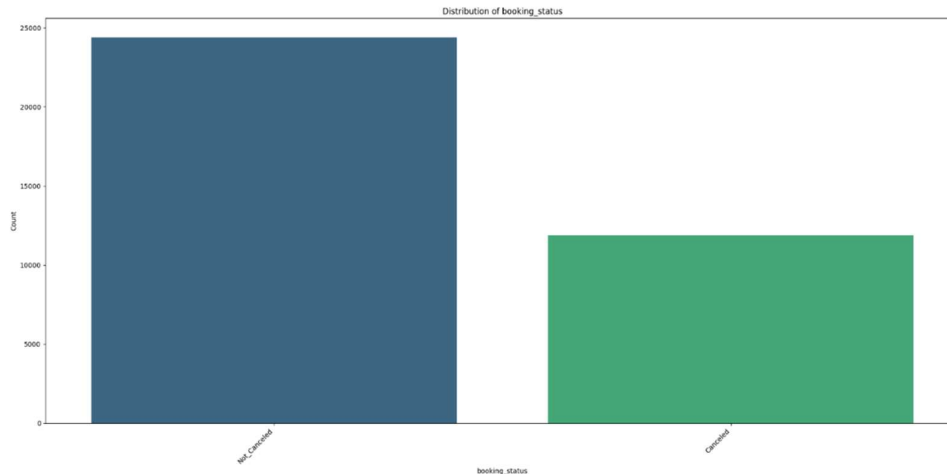


Figure 18 Univariate Booking Status

IV.3.2.5. Inferences from Univariate Analysis For Categorical Columns

type_of_meal_plan: The vast majority of guests opt for 'Meal Plan 1' (Breakfast), indicating a strong preference for a basic meal offering. 'Not Selected' is the second most common, suggesting some guests prefer flexibility or external dining, while 'Meal Plan 3' (Full Board) is rarely chosen.

room_type_reserved: 'Room_Type 1' is overwhelmingly the most frequently booked room type, followed by 'Room_Type 4'. This indicates that these two room types are either the most available, most affordable, or most desired by guests, potentially warranting closer examination of their features and pricing strategies.

market_segment_type: Online channels are the dominant source of bookings, highlighting the critical importance of the hotel's digital presence and online travel agency partnerships. Offline and Corporate segments also contribute significantly, while Aviation and Complementary bookings are minimal niche markets.

booking_status: The data shows that a larger proportion of bookings are 'Not_Canceled' compared to 'Canceled'. While this indicates a generally stable booking trend, the

percentage of 'Canceled' bookings is still significant enough to warrant further investigation into the reasons behind these cancellations to mitigate potential revenue loss.

IV.4. BiVariate Analysis

IV.4.1. Correlation Heatmap of Numerical Columns

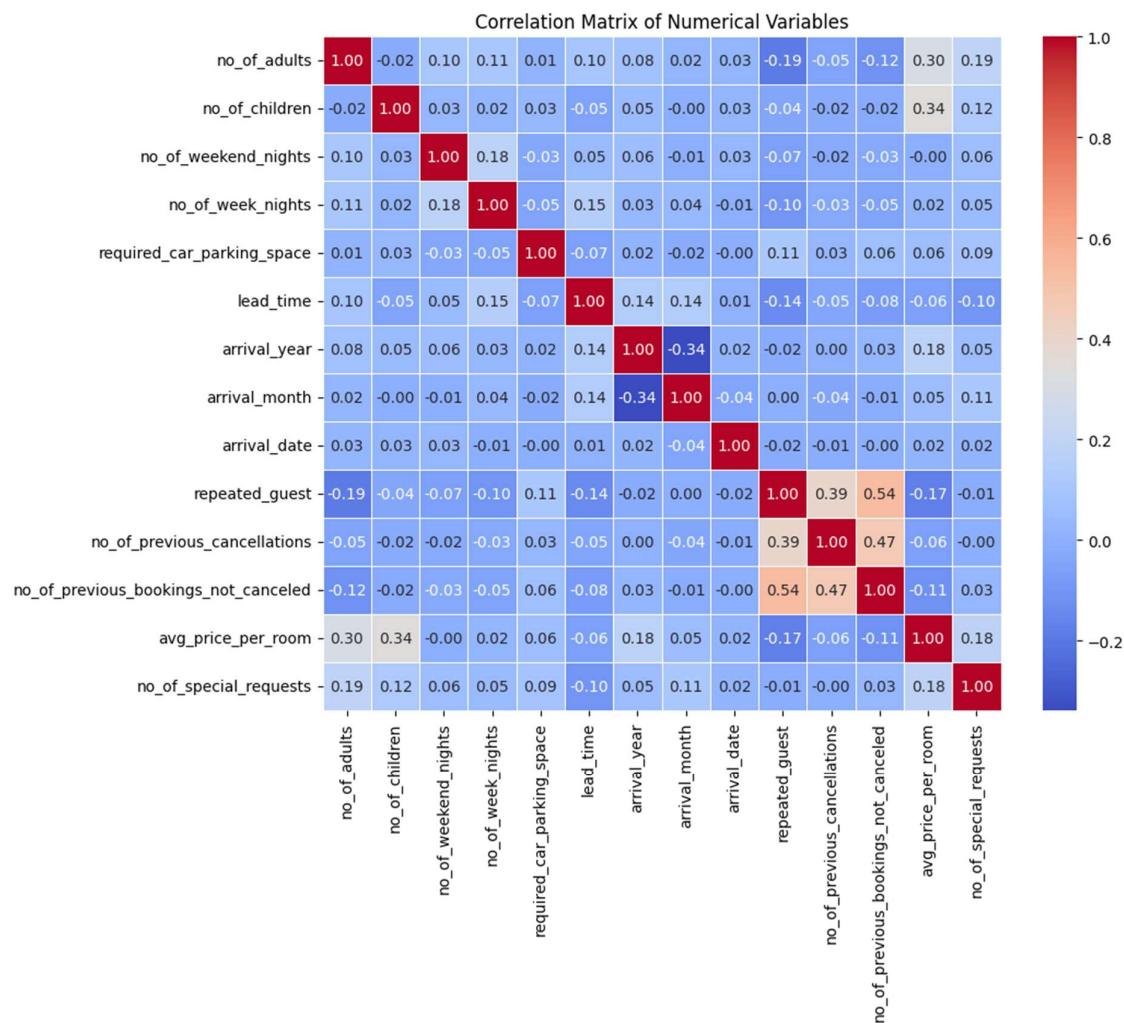


Figure 19 Correlation Map- Bivariate

IV.4.2. Bivariate Analysis – Numerical Vs Categorical

IV.4.2.1. Booking Status- Distribution

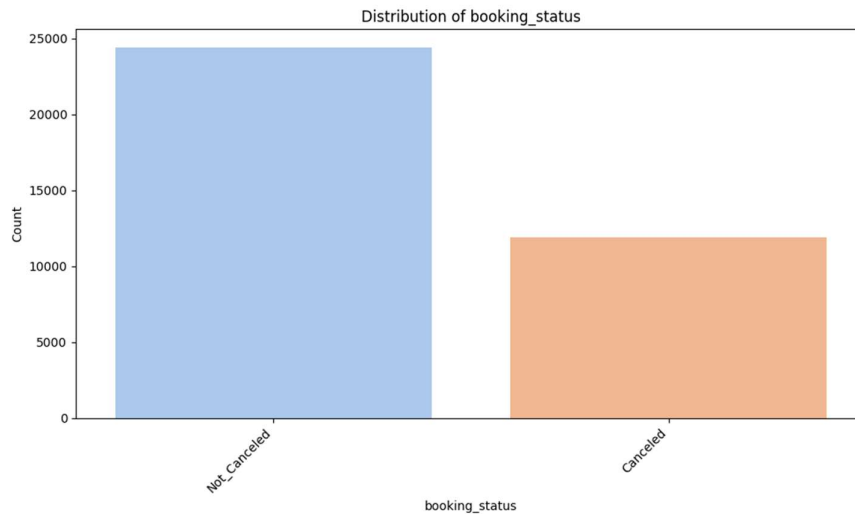


Figure 20 Booking Status

IV.4.2.2. No of Adults Versus Booking Status

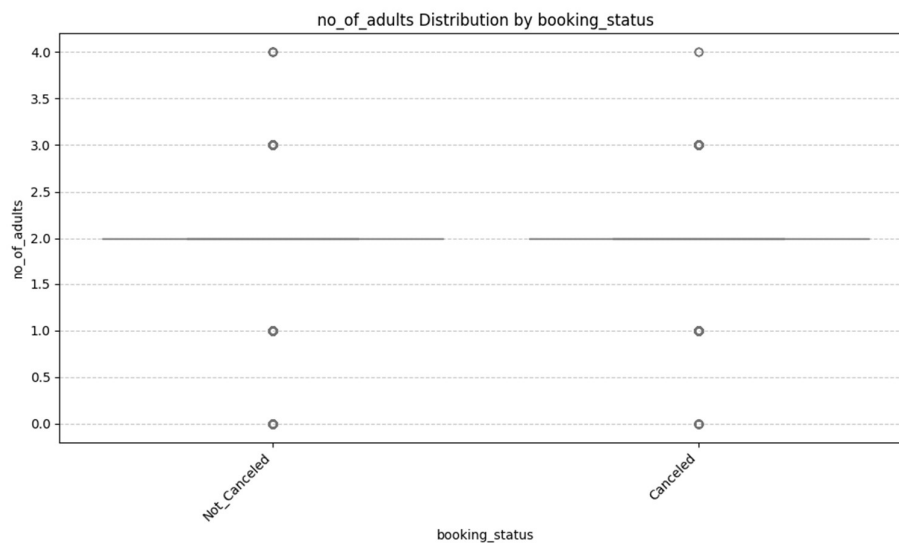


Figure 21 Bivariate Adults Vs Booking Status

IV.4.2.3. No of Children Versus Booking Status

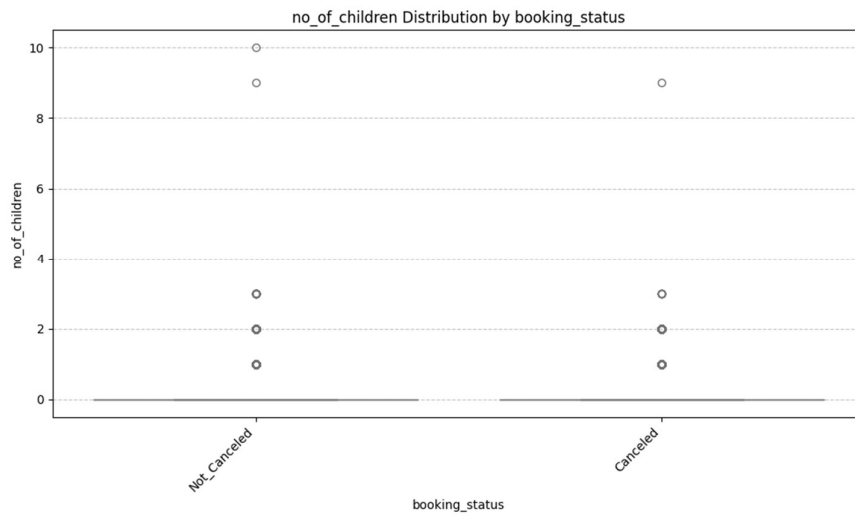


Figure 22 Bivaraiate Children vs Booking Status

IV.4.2.4. No of Weekend Nights Versus Booking Status

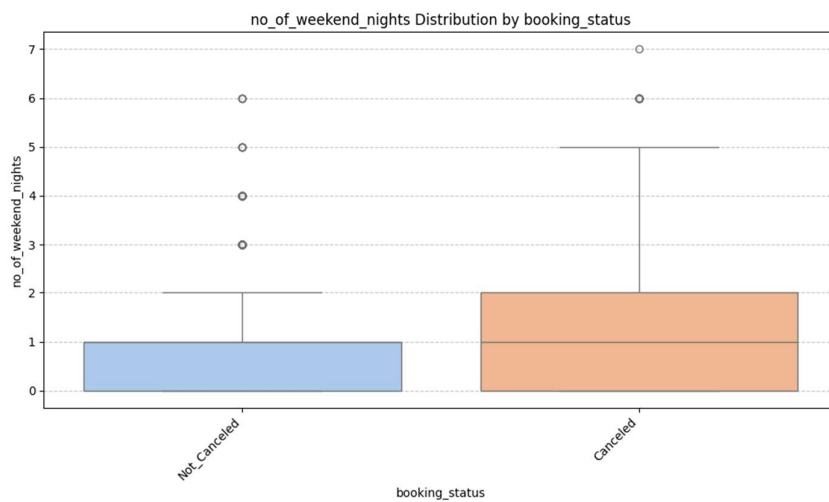


Figure 23 Bivariate Weekend Nights Vs Booking Status

IV.4.2.5. No of Weekday Nights Versus Booking Status

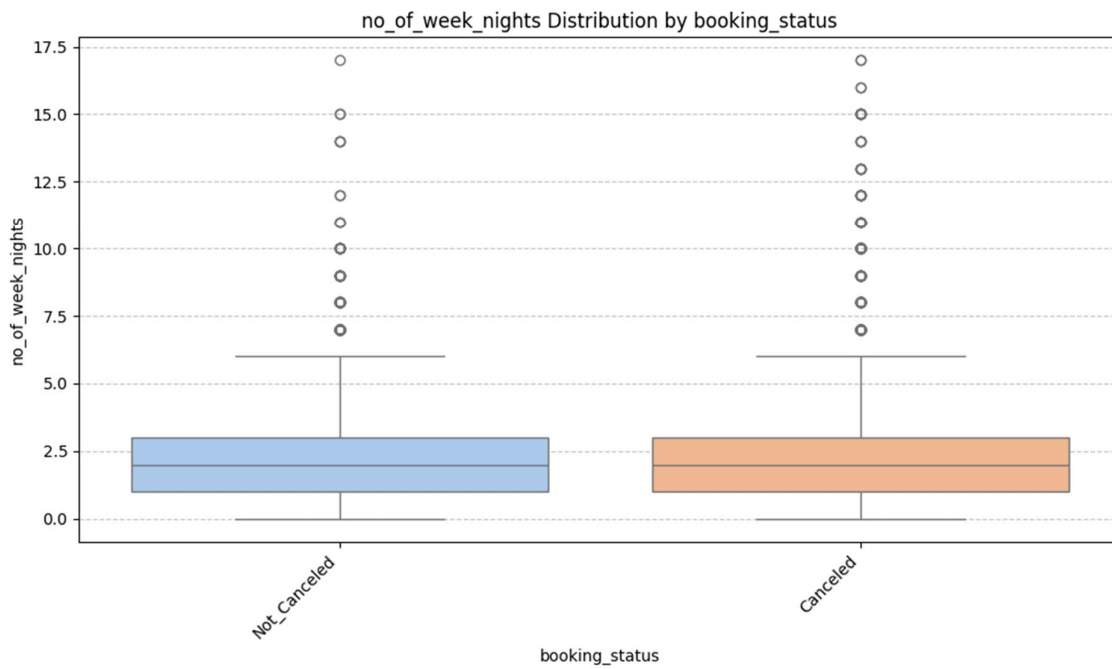


Figure 24 Bivariate Week Nights Vs Booking Status

IV.4.2.6. No of car parking space Versus Booking Status

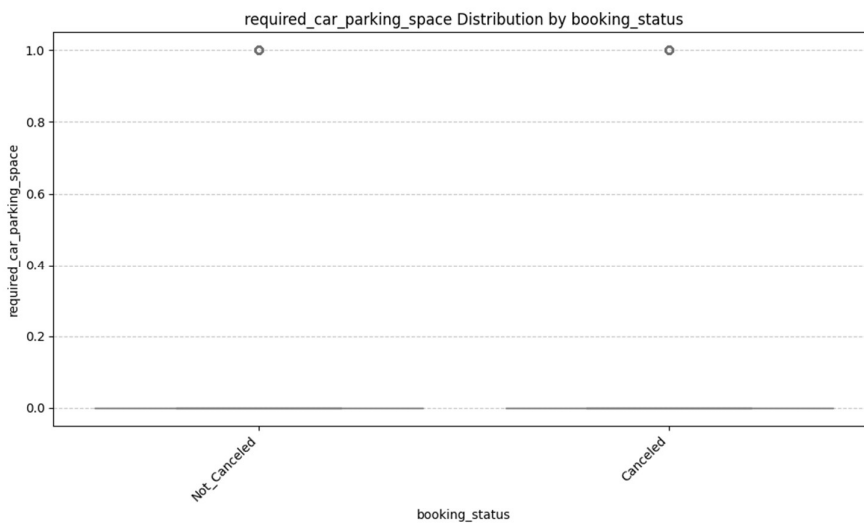


Figure 25 Bivariate Car Parking Need Vs Booking Status

IV.4.2.7. Lead Time Versus Booking Status

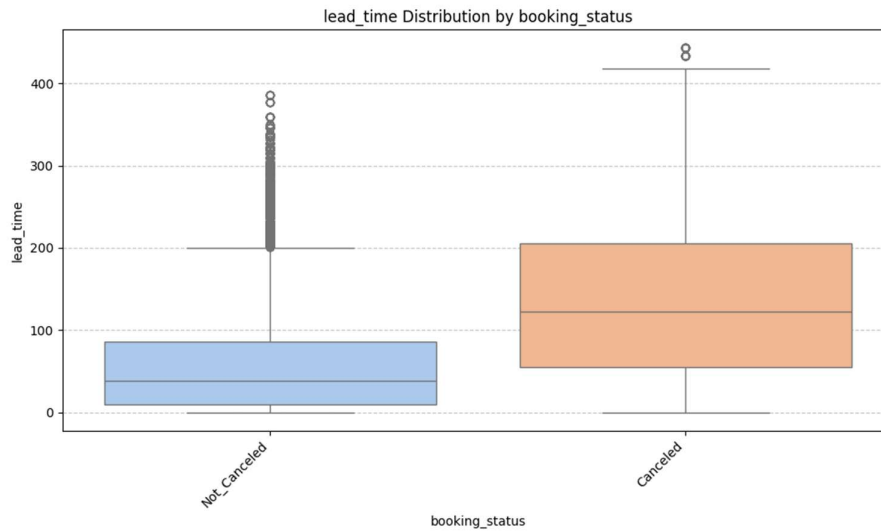


Figure 26 Bivariate Lead Time Vs Booking Status

IV.4.2.8. Arrival Year Versus Booking Status

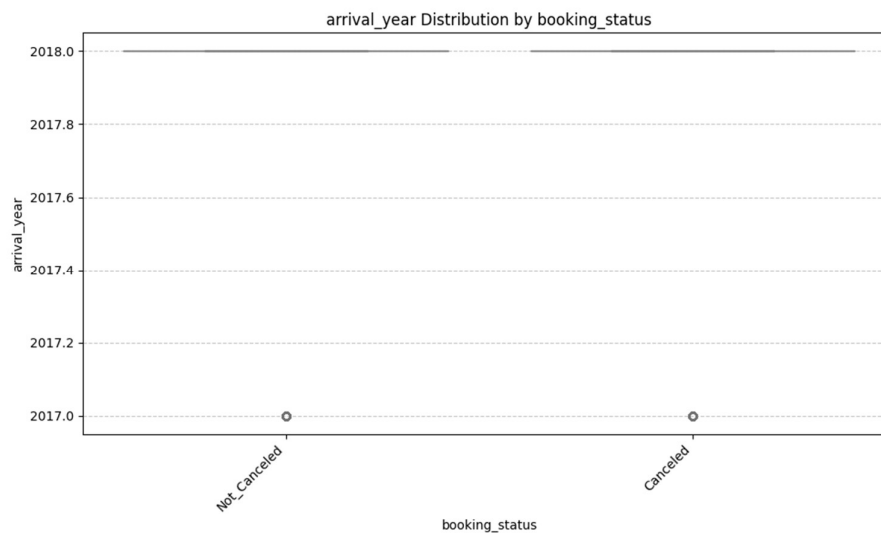


Figure 27 Bivariate Arrival Year Vs Booking Status

IV.4.2.9. Arrival Month Versus Booking Status

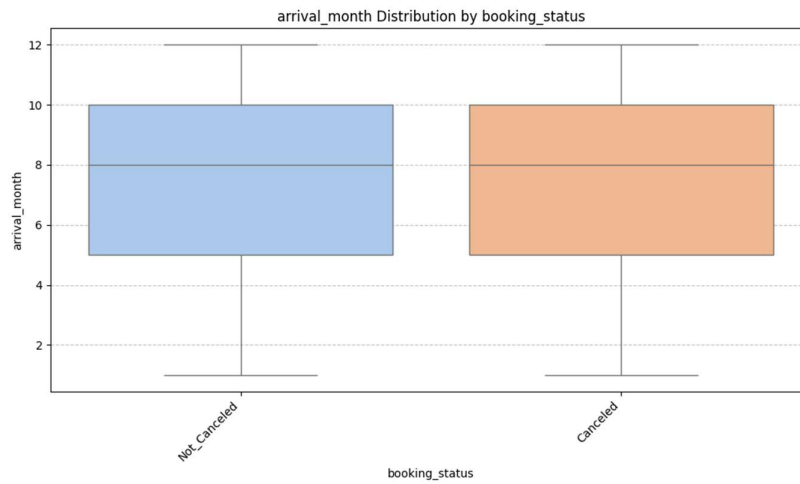


Figure 28 Bivariate Arrival Month Vs Booking Status

IV.4.2.10. Arrival Date Versus Booking Status

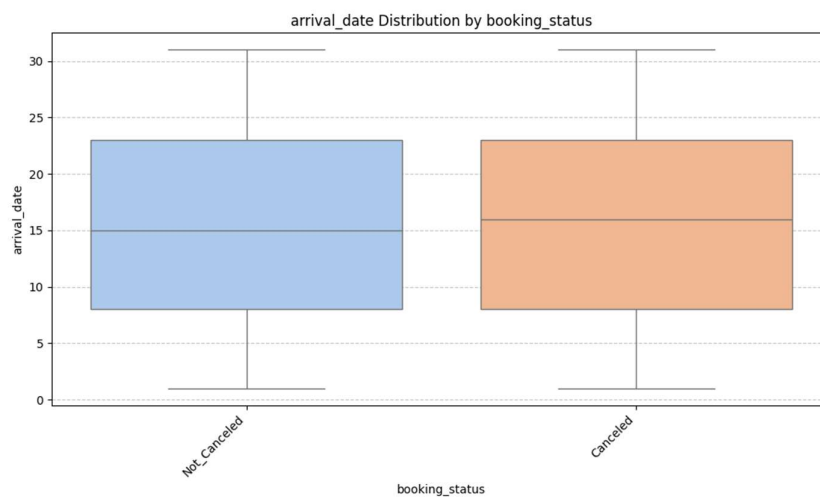


Figure 29 Bivariate Arriva Date Vs Booking Status

IV.4.2.11. Repeated Guest Versus Booking Status

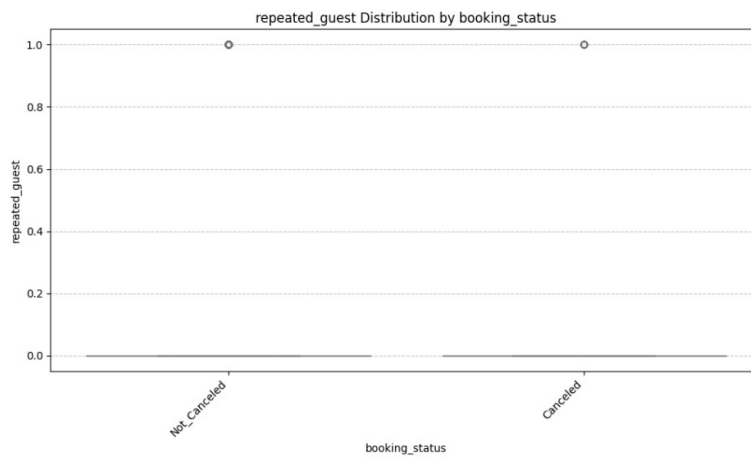


Figure 30 Bivariate Repeated Guest Vs Booking Status

IV.4.2.12. No of Previous Cancellations Versus Booking Status

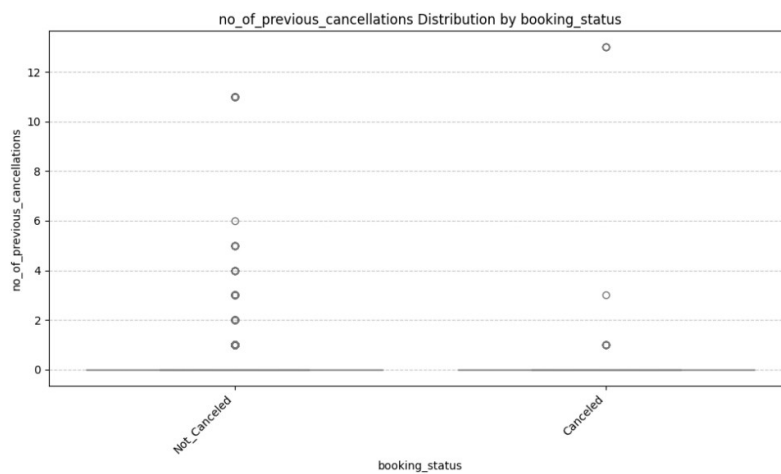


Figure 31 Bivariate Prev Cancellation Vs Booking Status

IV.4.2.13. No of Previous Bookings Not Cancelled Versus Booking Status

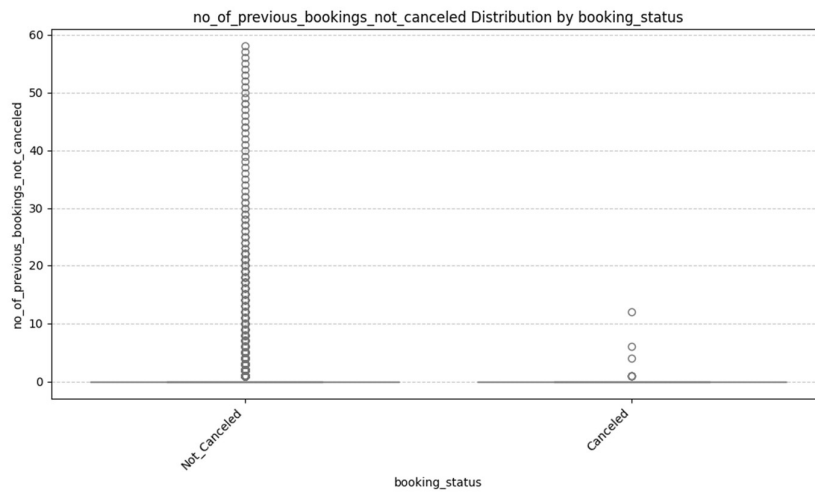


Figure 32 Bivariate Prev Booking Not Cancelled Vs Booking Status

IV.4.2.14. Average Price Room Versus Booking Status

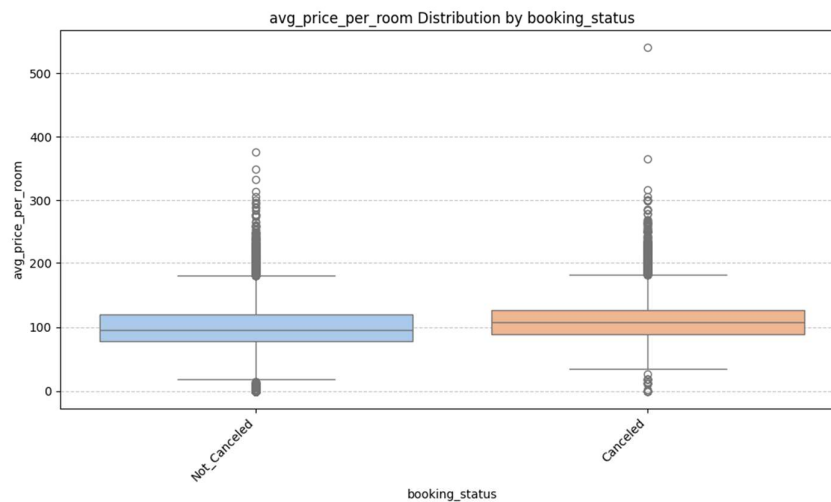


Figure 33 Bivariate Room Price Vs Booking Status

IV.4.2.15. No of Special Requests Versus Booking Status

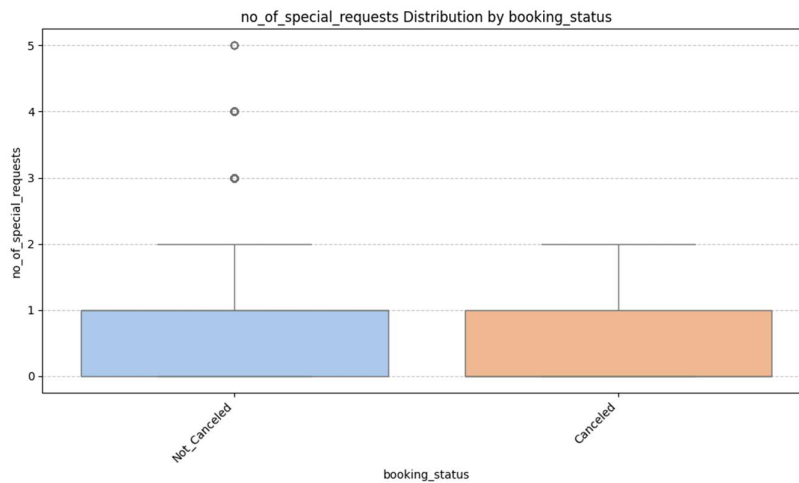


Figure 34 Bivariate Special Requests Vs Booking Status

IV.4.3. Inferences from Bivariate Analysis (Numerical vs. Booking Status)

Based on the box plots comparing numerical variables against booking_status, we can draw the following general inferences:

1. **Lead Time as a Predictor:** Bookings with a longer lead_time (time between booking and arrival) appear to have a higher propensity for cancellation. The box plot for lead_time likely shows that 'Canceled' bookings have a higher median and/or a larger interquartile range (IQR) than 'Not_Canceled' bookings.
2. **Average Price per Room (avg_price_per_room):** The distribution of avg_price_per_room might vary significantly between canceled and non-canceled bookings. For instance, bookings with very low or very high average prices might have different cancellation patterns, indicating price sensitivity or a different segment of customers.
3. **Special Requests and Commitment:** A higher no_of_special_requests often correlates with a lower cancellation rate. Guests who take the time to make special requests are generally more committed to their booking, suggesting a lower median for 'Canceled' bookings in this category.
4. **Previous Booking History:** Features such as no_of_previous_cancellations and repeated_guest are strong indicators. 'Canceled' bookings will show a higher concentration of previous cancellations, while

'repeated_guest' will predominantly be associated with 'Not_Canceled' bookings, highlighting customer loyalty.

5. **Room Occupancy and Stay Duration:** Variables

like no_of_adults, no_of_children, no_of_weekend_nights, and no_of_week_nights might show some differences, but often their distributions between 'Canceled' and 'Not_Canceled' bookings have substantial overlap, implying they might not be the primary standalone predictors of cancellation, but could contribute when combined with other factors.

IV.4.4. Bivariate Analysis of Categorical Vs Target Column

Categorical columns:

type_of_meal_plan

room_type_reserved

market_segment_type

Versus

booking_status

IV.4.4.1. Cross-tabulation for booking_status by type_of_meal_plan:

Table 1 Bivariate - Booking Status Vs Meal Plan

booking_status	Canceled	Not_Canceled
type_of_meal_plan		
Meal Plan 1	8679	19156
Meal Plan 2	1506	1799
Meal Plan 3	1	4
Not Selected	1699	3431

cal Columns ---

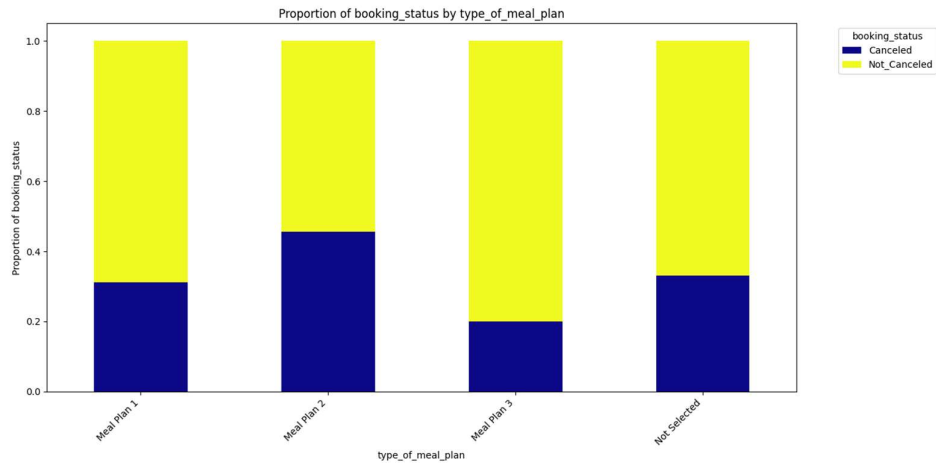


Figure 35 Bivariate Meal Plan Vs Booking Status

IV.4.4.2. Cross-tabulation for booking_status by room_type_reserved:

Table 2 Bivariate Booking Status VS Room Type

booking_status	Canceled	Not_Canceled
room_type_reserved		
Room_Type 1	9072	19058
Room_Type 2	228	464
Room_Type 3	2	5
Room_Type 4	2069	3988
Room_Type 5	72	193
Room_Type 6	406	560
Room_Type 7	36	122

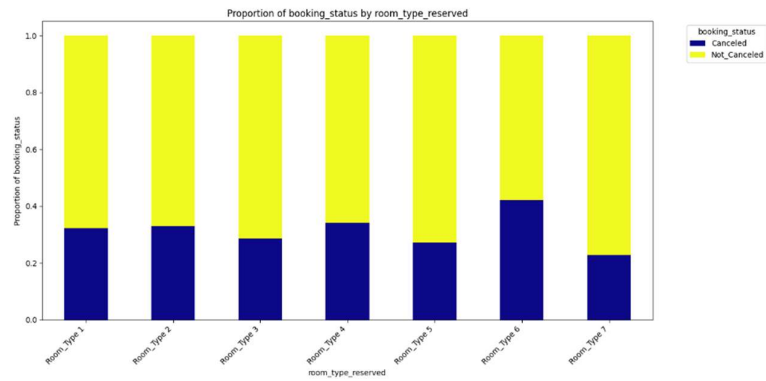


Figure 36 Bivariate Booking Status VS Room Type

IV.4.4.3. Cross-tabulation for booking_status by market_segment_type:

Table 3 Bivariate Market Segment Type VS Booking Status

booking_status	Canceled	Not_Canceled
market_segment_type		
Aviation	37	88
Complementary	0	391
Corporate	220	1797
Offline	3153	7375
Online	8475	14739

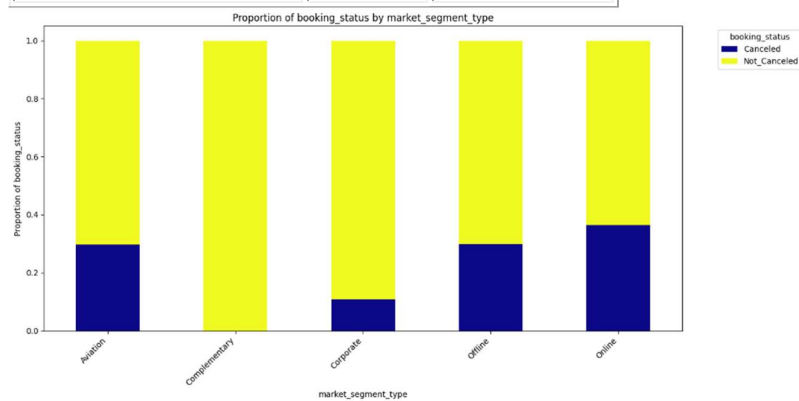


Figure 37 Bivariate Market Segment Type VS Booking Status

IV.5. Answers to EDA Questions

IV.5.1. What are the busiest months in the hotel?

We analyse the bookings per month for finding the busiest months.

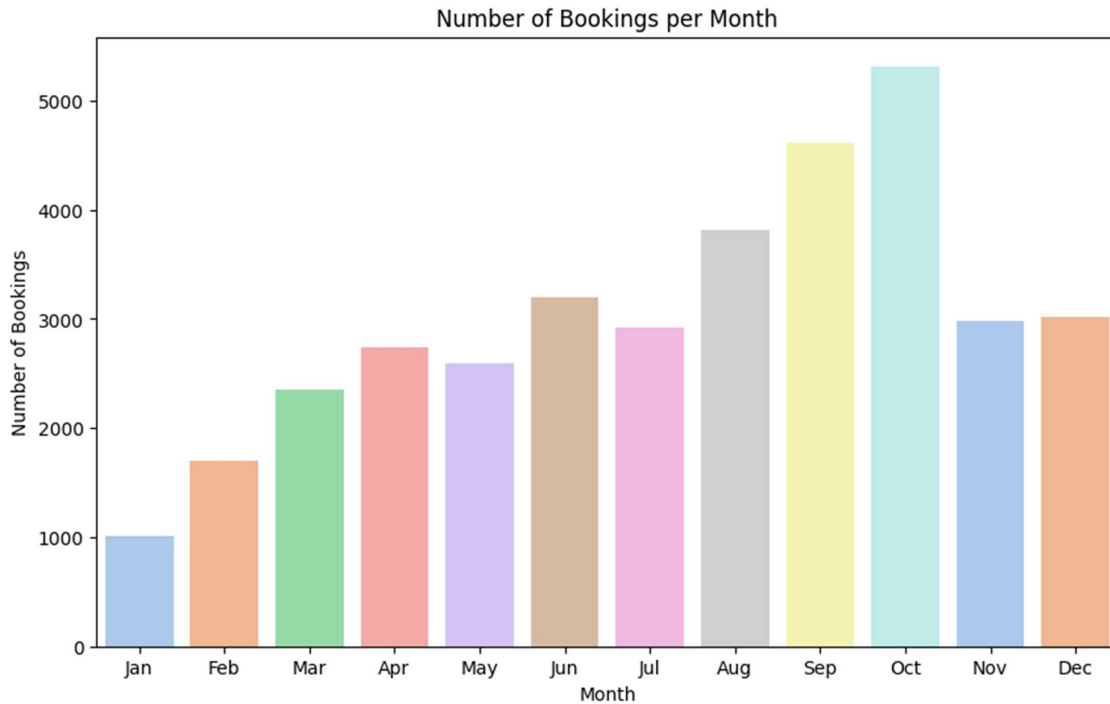


Figure 38 EDA - Busy Months

The busiest months in the hotel based on the number of bookings are:

arrival_month	No of Bookings
10	5317
9	4611
8	3813
Answer : The busiest month in the hotel is October . It is followed by September and August. The lowest busy month is January	

IV.5.2. Which market segment do most of the guests come from?

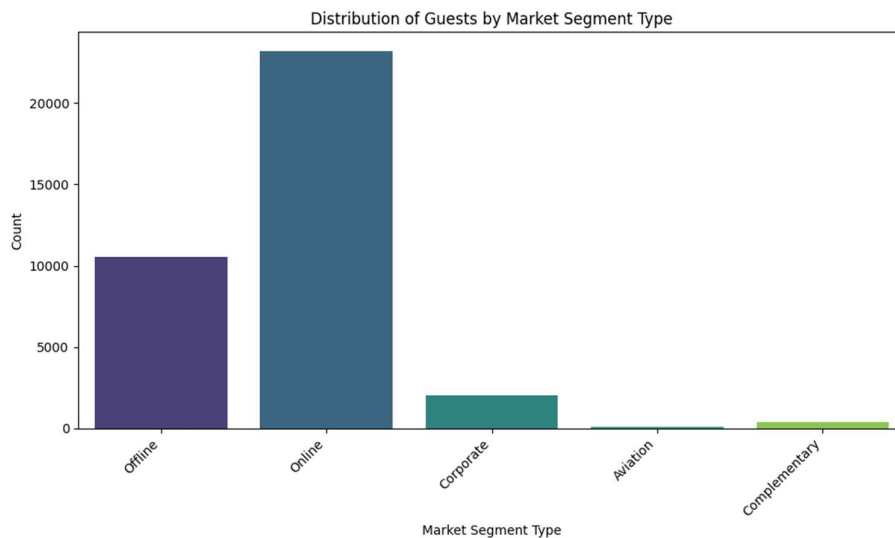


Figure 39 EDA - Guests By Market segment Type

Answer:

Most of the guests come from the **Online** market segment, accounting for 23214 bookings.

The distribution of market segments is as follows:

- **Online:** 23214 bookings
- **Offline:** 10528 bookings
- **Corporate:** 2017 bookings
- **Complementary:** 391 bookings
- **Aviation:** 125 bookings

IV.5.3. Hotel rates are dynamic and change according to demand and customer demographics. What are the differences in room prices in different market segments?

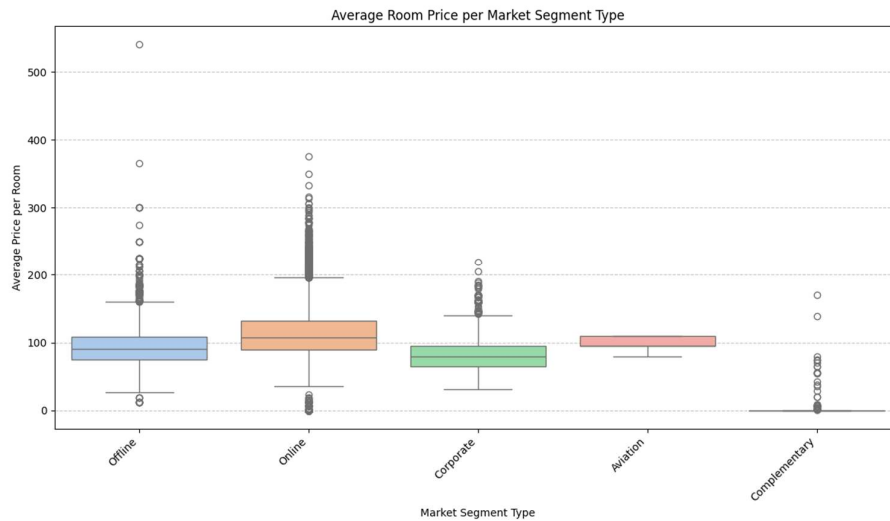


Figure 40 EDA - Room Price in Market Segments

Answer : Inferences from Average Room Price per Market Segment Type are as below:

- **Online Segment:** This segment generally exhibits the highest median average prices and the widest range, indicating dynamic pricing and high demand through online booking channels.
- **Offline Segment:** While still showing a broad price range, the median average price for offline bookings appears slightly lower than online, suggesting a different pricing structure or customer base.
- **Corporate Segment:** Bookings from the corporate segment tend to have a more consistent and often lower average price, likely due to pre-negotiated rates or bulk bookings.
- **Complementary Segment:** This segment shows the lowest average prices, with a significant number of bookings potentially at zero cost, indicative of promotional stays or special guest arrangements.
- **Aviation Segment:** This segment has a moderate average price range, often with a tighter distribution, which might reflect specific contractual agreements or a specialized clientele.

Summary: The pricing strategy for INN Hotels is highly dynamic and segment-dependent, with online channels driving the highest average prices, while corporate and complementary segments often benefit from lower or special rates.

IV.5.4. What percentage of bookings are cancelled?

To analyse this, find the no of booking with Canceled status and then calculate the percentage.



Figure 41 EDA - Booking Cancelled OR Not Cancelled

Answer:

Total bookings: 36275

Canceled bookings: 11885

Percentage of canceled bookings: **32.76%**

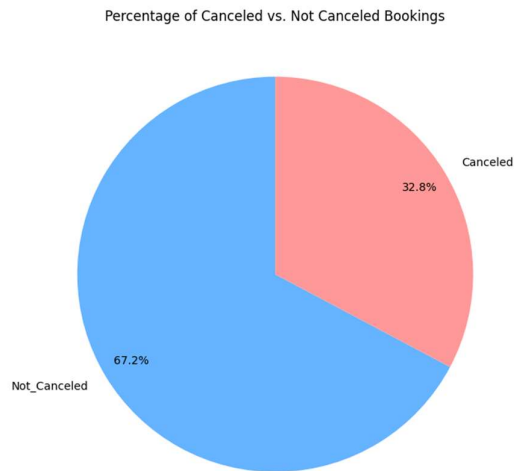


Figure 42 Pie Chart - Booking Cancelled OR Not Cancelled

IV.5.5. Repeating guests are the guests who stay in the hotel often and are important to brand equity. What percentage of repeating guests cancel?

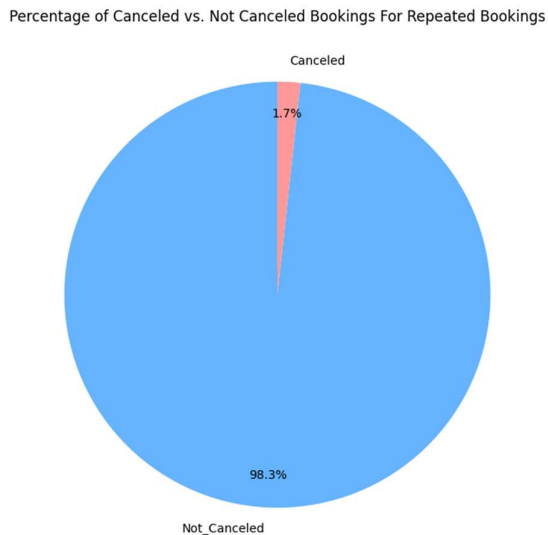


Figure 43 Pie Chart - Percent of Repeated Guests Who Cancel

Answer:

Total repeating guest bookings: 930

Canceled repeating guest bookings: 16

Percentage of repeating guests who cancel: 1.72%

This indicates that a very small percentage of repeating guests cancel their bookings, highlighting the loyalty and reliability of repeated guests who comes.

IV.5.6. Many guests have special requirements when booking a hotel room. Do these requirements affect booking cancellation?

Descriptive Statistics for no of special requests by Booking Status:

	count	mean	std	min	25%	50%	75%	max
booking_status								
Canceled	11885.0	0.334623	0.574345	0.0	0.0	0.0	1.0	2.0
Not_Canceled	24390.0	0.758549	0.836532	0.0	0.0	1.0	1.0	5.0

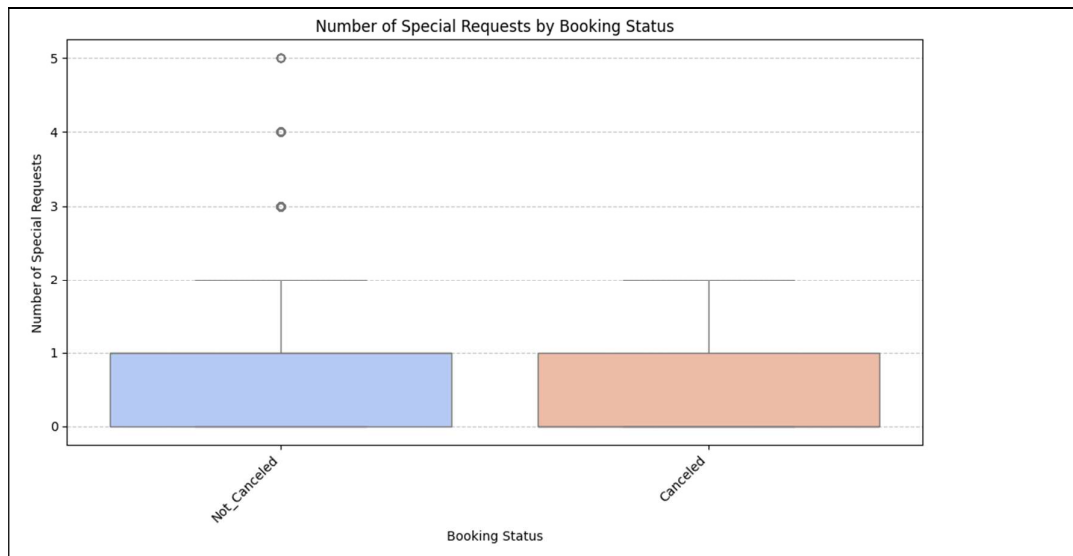


Figure 44 EDA - Special Requests Vs Booking Status

Canceled Bookings: The mean number of special requests is approximately 0.33, with a median (50th percentile) of 0. This indicates that most canceled bookings had no special requests, or very few.

Not Canceled Bookings: The mean number of special requests is significantly higher at approximately 0.76, with a median (50th percentile) of 1. This suggests that non-canceled bookings tend to have more special requests.

Inference: There is a clear difference in the number of special requests between canceled and not-canceled bookings. Bookings with special requests are less likely to be canceled. This could imply that guests who make special requests are more committed to their reservations, as they have invested more effort into planning their stay. This information is valuable for understanding guest commitment and potentially for identifying bookings that are at a lower risk of cancellation.

Customer Commitment: This suggests that guests who take the time to make special requests are generally more committed to their booking and are less likely to cancel. These requests could indicate a higher investment in the trip planning, making cancellation less desirable for the guest.

Operational Implications: This observation could be valuable for hotels, as bookings with special requests might represent a more reliable revenue stream. Hotels could potentially prioritize fulfilling these requests to further enhance guest satisfaction and reduce the already low cancellation risk from this segment.

IV.6. Key meaningful observations on individual variables and the relationship between variables

General Booking Patterns:

- **Occupancy:** Most bookings are for 2 adults. Bookings with children are rare (75% have 0 children).
- **Stay Duration:** Stays are generally short, typically 1-3 weeknights and up to 2 weekend nights.
- **Parking:** Very few guests (around 3%) require a car parking space.
- **Lead Time:** Bookings exhibit a wide lead time, with a median of 57 days, indicating both last-minute and planned reservations.

- **Arrival Trend:** The data is concentrated in 2017 and 2018. October, September, and August are identified as the busiest months for arrivals.

Guest Loyalty and Cancellation Behavior:

- **Repeat Guests:** The hotel has a very low percentage of repeated guests (around 2.5%).
- **Previous History:** Most guests have no prior cancellations or non-canceled bookings, suggesting a largely new customer base.
- **Cancellation Rate:** The overall booking cancellation rate is significant, standing at **32.76%**.
- **Repeat Guest Loyalty:** Repeating guests show exceptionally high loyalty, with only **1.72%** of their bookings being canceled, indicating they are a highly reliable segment.

Market Segments and Pricing:

- **Dominant Market:** The 'Online' market segment is by far the most dominant, accounting for the majority of bookings (over 23,000).
- **Dynamic Pricing:** The average price per room is highly dynamic, ranging from 0 to 540 euros. 'Online' bookings generally command the highest average prices, while 'Corporate' and 'Complementary' segments tend to have lower or special rates.

Factors Influencing Cancellation:

- **Lead Time:** Bookings made with a longer lead_time tend to have a higher propensity for cancellation.
- **Special Requests:** A crucial finding is the inverse relationship between special requests and cancellation. Bookings with more special requests are significantly less likely to be canceled (mean special requests: 0.76 for Not_Canceled vs. 0.33 for Canceled). This suggests that guests who invest effort in making specific requests are more committed to their stay.
- **Previous Booking History:** repeated_guest and no_of_previous_cancellations are strong indicators; repeat guests and those with no prior cancellations are less likely to cancel the current booking.

Relationships Between Variables:

- **Correlation:** There is a positive correlation (0.54) between repeated_guest and no_of_previous_bookings_not_canceled, which is expected as repeat guests are those with previous non-canceled bookings.

- **Price and Occupancy:** Moderate positive correlations exist between no_of_children and avg_price_per_room (0.34), and no_of_adults and avg_price_per_room (0.30). This suggests that family-friendly or larger group bookings often correspond to higher average room prices.

V. Data Preprocessing

V.1. Missing Value Treatment

There are no missing values in data and hence no missing value treatment is required

There are 36275 rows and 19 columns. There are no duplicate ones.

V.2. Outlier Detection and Treatment

Box Plots to check the outliers:

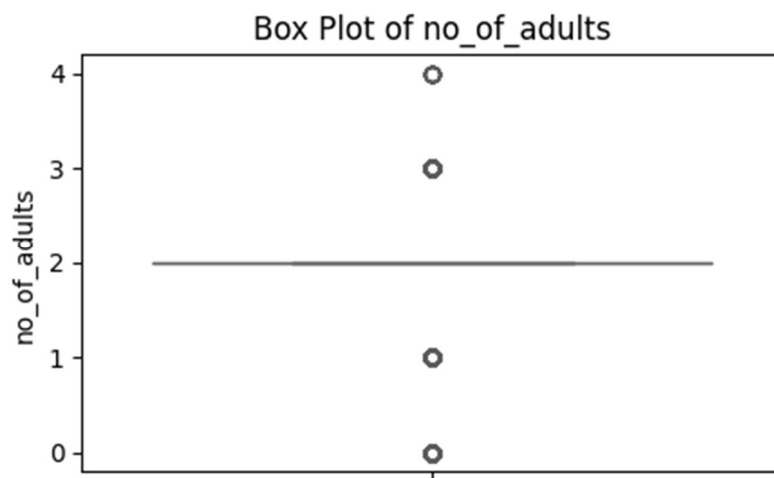


Figure 45 Outlier Check - Box Plot For Adults

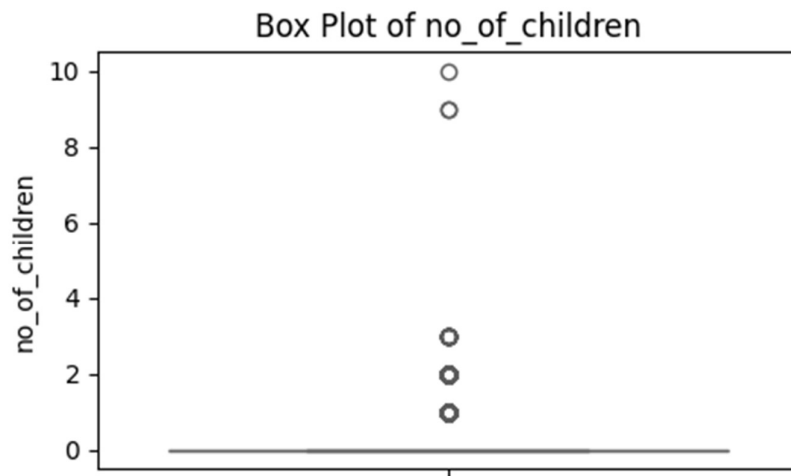


Figure 46 Outlier Check - Box Plot For Children

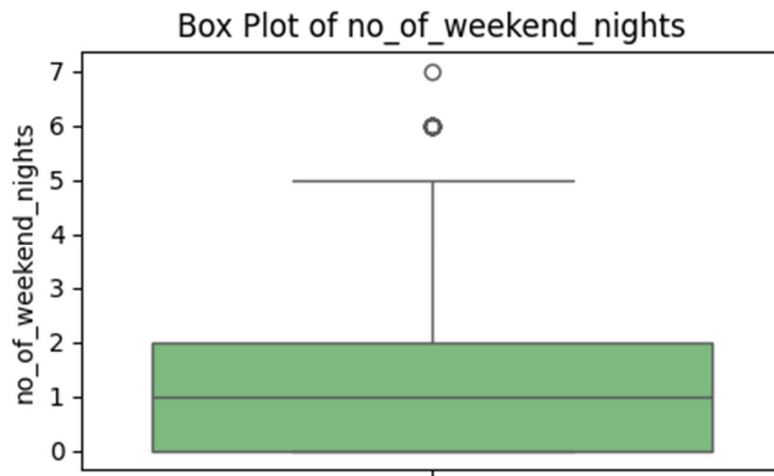


Figure 47 Outlier Check - Box Plot For Weekend Nights

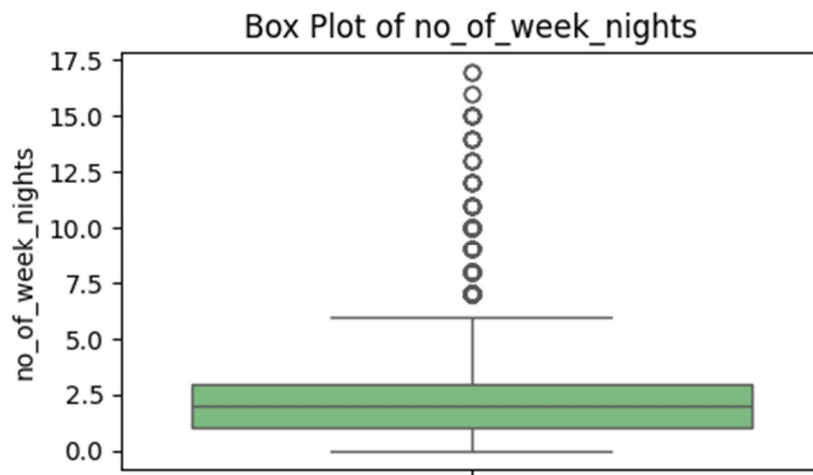


Figure 48 Outlier Check - Box Plot For Week Nights

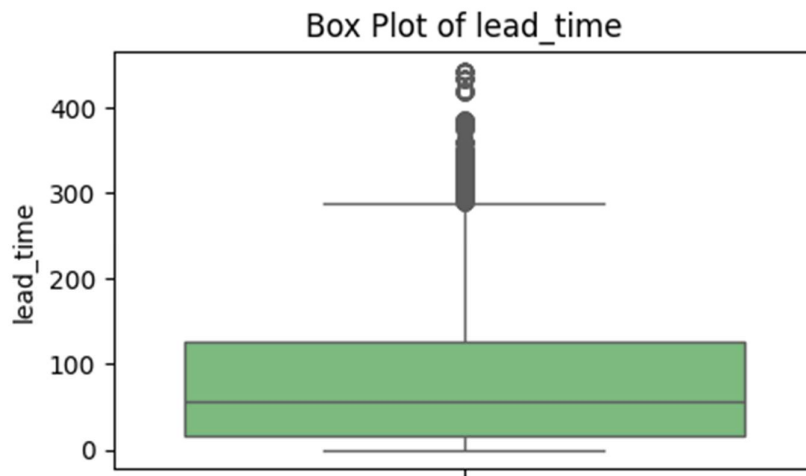


Figure 49 Outlier Check - Box Plot For Lead Time

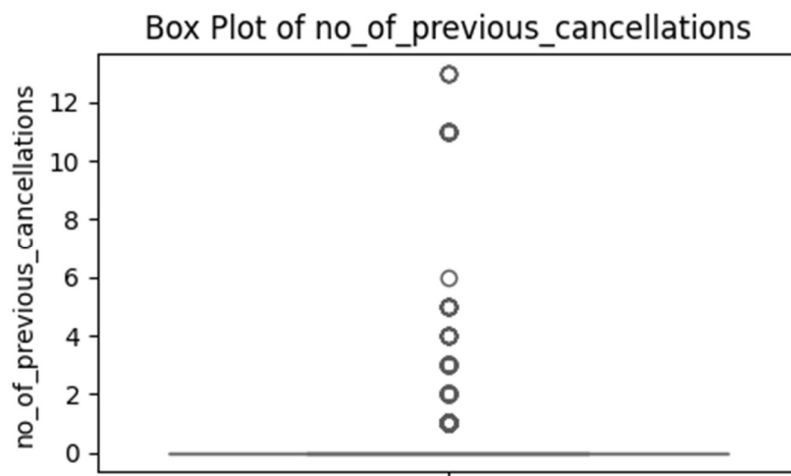


Figure 50 Outlier Check - Box Plot For No of Prev Cancellations

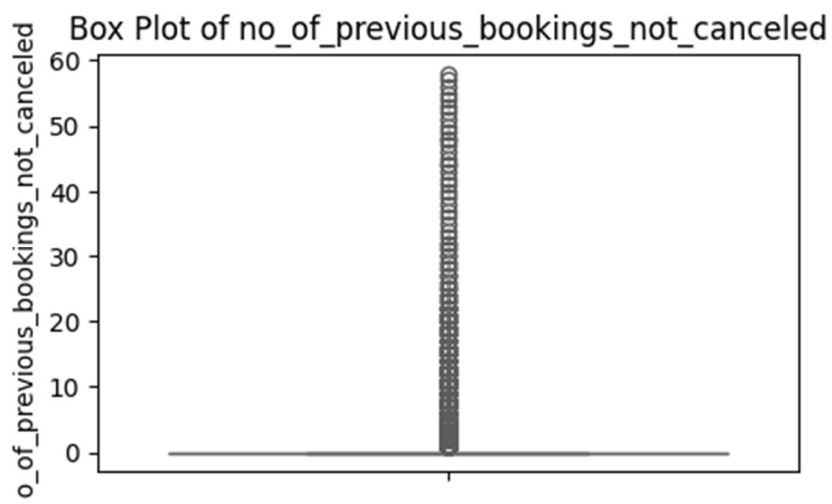


Figure 51 Outlier Check - Box Plot For Prev Booking Not Cancelled

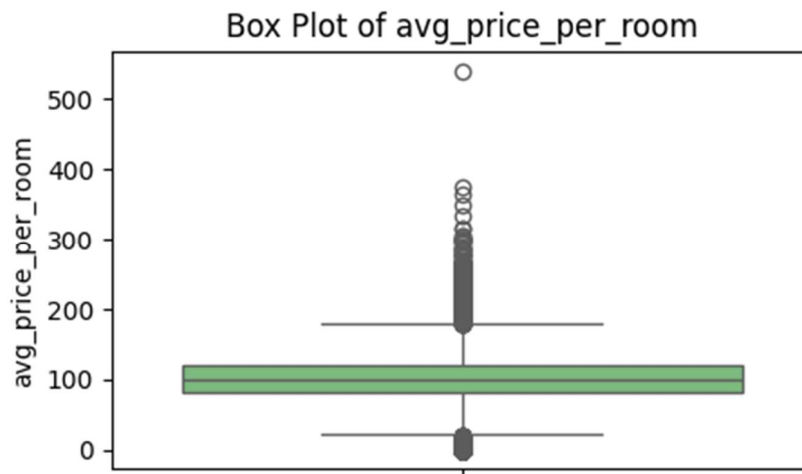


Figure 52 Outlier Check - Box Plot For Avg Room Price

Based on the box plots generated for the numerical columns, we can observe the following regarding outliers:

no_of_adults: While most bookings are for 2 adults, there are some outliers where no_of_adults is 0 (which might indicate data entry errors or very specific booking types, e.g., only children), and also a few bookings with 3 or 4 adults.

no_of_children: The majority of bookings have 0 children. There are clear outliers where the number of children is 1, 2, 3, and even up to 10, indicating family bookings.

no_of_weekend_nights: Most bookings involve 0, 1, or 2 weekend nights. Outliers are present for stays involving 3 or more weekend nights, extending up to 7 nights.

no_of_week_nights: Similar to weekend nights, the bulk of bookings are for 1, 2, or 3 weeknights. There are significant outliers for longer stays, reaching up to 17 weeknights.

lead_time: The lead_time distribution shows a long tail of outliers, with many bookings made far in advance (over 200 days, up to 443 days), while the majority are made within a shorter period.

no_of_previous_cancellations: Overwhelmingly, guests have no previous cancellations. The box plot clearly shows outliers for guests who have 1, 2, or more previous cancellations (up to 13), indicating a small group of repeat cancellers.

no_of_previous_bookings_not_canceled: Most guests have 0 previous non-canceled bookings. Outliers exist for guests with multiple previous successful bookings (e.g., 1, 2, and up to 58), representing loyal, repeated guests.

avg_price_per_room: The average price per room has outliers at the lower end (including 0, which could be promotional or internal bookings) and at the higher end, indicating premium room types or high-demand periods (up to 540 euros).

These outliers suggest a diverse customer base and booking behaviors, and their treatment (e.g., capping) helps to prevent extreme values from disproportionately influencing model training later on.

If we remove these outliers, it might affect the booking behaviours captured. It is better to handle it in model.

Decision Tree Classifier (Robust by Design): Decision Trees are non-parametric and split data based on thresholds. The "extreme" nature of an outlier doesn't pull the split point the way it pulls a mean in linear models and hence no treatment is needed for the Decision Tree model.

Logistic Regression (Sensitivity Check): Unlike trees, Logistic Regression is a mathematical model that can be influenced by the magnitude of outliers. To fix this simply swap `StandardScaler` for `RobustScaler` (`sklearn.preprocessing.RobustScaler`). It scales data based on the Median and Interquartile Range (IQR) rather than the Mean and Standard Deviation. This ensures that the extreme values don't "shrink" the rest of your data into a tiny range.

V.3. Feature Engineering

Duplicate the data frame

1. For the Logistic Regression
2. For the Decision Tree

and then do the feature engineering

V.3.1. Feature Engineering For Logistic Regression

1. Remove the Unique Identified for Booking ID
2. Target Variable To be Converted to Numerical

Convert the `booking_status` column into numerical format (e.g., 'Not_Canceled' to 0 and 'Canceled' to 1).

3. One Hot Encoding:

Convert categorical features (`type_of_meal_plan`, `room_type_reserved`, `market_segment_type`) into numerical representations

4. Numerical Feature Scaling: Scale numerical features (`no_of_adults`, `no_of_children`, `no_of_weekend_nights`, `no_of_week_nights`, `lead_time`, `avg_price_per_room`, `no_of_special_requests`, etc.). Since logistic regression is sensitive to outliers and the

descriptive text mentioned using RobustScaler, this would be a suitable choice. RobustScaler is less affected by outliers than StandardScaler.

5. Feature Transformation (for skewed data): Apply transformations like log1p to highly skewed numerical features (e.g., lead_time, avg_price_per_room, no_of_previous_cancellations, no_of_previous_bookings_not_canceled) to make their distributions more Gaussian-like and reduce the impact of extreme values, which can improve model performance.
6. Date-related Features: Although arrival_year, arrival_month, and arrival_date are already numerical, we consider combining them into a single date feature and extracting other time-based features if relevant (e.g., day of week, season).
7. One Hot Encoding For Season:

Convert season into numerical representations

8. Drop the arrival_full_date created to figure out whether it is weekend or weekday.
9. Drop the season object field after one hot encoding

The columns after Feature Engineering in this case are as below:

Data columns (total 31 columns)

```
['no_of_adults', 'no_of_children', 'no_of_weekend_nights', 'no_of_week_nights',  
'required_car_parking_space', 'lead_time', 'arrival_year', 'repeated_guest',  
'no_of_previous_cancellations', 'no_of_previous_bookings_not_canceled',  
'avg_price_per_room', 'no_of_special_requests', 'booking_status',  
'type_of_meal_plan_Meal Plan 2', 'type_of_meal_plan_Meal Plan 3',  
'type_of_meal_plan_Not Selected', 'room_type_reserved_Room_Type 2',  
'room_type_reserved_Room_Type 3', 'room_type_reserved_Room_Type 4',  
'room_type_reserved_Room_Type 5', 'room_type_reserved_Room_Type 6',  
'room_type_reserved_Room_Type 7', 'market_segment_type_Complementary',  
'market_segment_type_Corporate', 'market_segment_type_Offline',  
'market_segment_type_Online', 'is_weekend', 'is_complimentary', 'season_Spring',  
'season_Summer', 'season_Winter']
```

V.3.2. Feature Engineering For Decision Tree

1. Remove the Unique Identified for Booking ID
2. Target Variable To be Converted to Numerical Convert the booking_status column into numerical format (e.g., 'Not_Canceled' to 0 and 'Canceled' to 1).

3. Label Encoding: Convert categorical features (type_of_meal_plan, room_type_reserved, market_segment_type) into numerical representations
4. Date-related Features: Although arrival_year, arrival_month, and arrival_date are already numerical, we consider combining them into a single date feature and extracting other time-based features if relevant (e.g., day of week, season).
5. Label Encoding For Season:

Convert season into numerical representations

6. Drop the arrival_full_date created to figure out whether it is weekend or weekday.

The columns after Feature Engineering in this case are as below:

Data columns (total 19 columns)

```
['no_of_adults', 'no_of_children', 'no_of_weekend_nights', 'no_of_week_nights',  
'type_of_meal_plan', 'required_car_parking_space', 'room_type_reserved', 'lead_time',  
'arrival_year', 'market_segment_type', 'repeated_guest', 'no_of_previous_cancellations',  
'no_of_previous_bookings_not_canceled', 'avg_price_per_room', 'no_of_special_requests',  
'booking_status', 'is_weekend', 'season', 'is_complimentary']
```

V.4. Data Scaling

Given the different sensitivities of Logistic Regression and Decision Trees to feature scales and outliers, we will employ distinct scaling strategies for Regression data Frame and Decision Tree Data Frame.

For Logistic Regression: Logistic Regression models are sensitive to the magnitude of features and outliers.:

1. **Skewed Feature Transformation:** Numerical features identified as highly skewed (e.g., lead_time, avg_price_per_room, no_of_previous_cancellations, no_of_previous_bookings_not_canceled) will undergo a log1p transformation. This helps to reduce skewness and make their distributions more Gaussian-like, which can improve model performance and prevent extreme values from dominating. Currently it is not done, after seeing and analysing the data
2. **Robust Scaling:** All numerical features will be scaled using RobustScaler. This scaler is preferred over StandardScaler because it scales data based on the median and interquartile range (IQR), making it more robust to outliers. This aligns with our observation of outliers in several numerical columns and the sensitivity of Logistic Regression.

For Decision Tree : Decision Trees and tree-based models, in general, are non-parametric and not sensitive to the scale of features. They split data based on thresholds, so the absolute magnitude of a feature does not affect how the splits are determined.

1. **No Scaling Required:** Feature scaling is typically not necessary for Decision Trees. The model will perform similarly whether features are scaled or not. However, for consistency or if we were to incorporate other models in an ensemble, a simple scaling method like StandardScaler could be applied, though it's not strictly mandatory for the Decision Tree itself.

Inference: By applying RobustScaler to regression data frame, we aim to mitigate the impact of outliers and skewed distributions, leading to a more stable and potentially better-performing Logistic Regression model. For decision tree data frame, we can proceed without extensive scaling due to the inherent robustness of Decision Trees to feature magnitudes.

V.5. Train-test split

V.5.1. Train test split For Logistic Regression:

Apply RobustScaler to the selected numerical columns [

```
'no_of_adults',  
'no_of_children',  
'no_of_weekend_nights',  
'no_of_week_nights',  
'lead_time',  
'arrival_year',  
'no_of_previous_cancellations',  
'no_of_previous_bookings_not_canceled',  
'avg_price_per_room',  
'no_of_special_requests'
```

]

in both training and testing sets

We have used **70:30 split**

There are 25366 rows in training data and 10872 rows in test data. Both of them are having 30 columns.

V.5.2. Train test split For Decision Tree:

We have used **70:30 split**.

There are 25366 rows in training data and 10872 rows in test data. Both of them are having 18 columns.

VI. Model Building

VI.1. Model Building For Logistic regression:

VI.1.1. Choosing the Metric to Optimize for the Problem

For the INN Hotels Group, the primary objective is to predict booking cancellations to **formulate profitable policies for cancellations and refunds** and **mitigate revenue loss**.

This is a binary classification problem where we are trying to predict if a booking will be 'Canceled' (positive class) or 'Not_Canceled' (negative class).

Let's consider the implications of different types of errors:

- **False Positives (Type I Error):** The model predicts a booking will be **Canceled**, but it is actually **Not_Canceled**.
 - **Business Impact:** The hotel might prematurely release the room, offer unnecessary incentives, or invest resources in trying to retain a guest who would have shown up anyway. This leads to wasted effort and potentially lost revenue if the room could have been sold at a higher price.
- **False Negatives (Type II Error):** The model predicts a booking will **Not_Canceled**, but it actually **Is Canceled**.
 - **Business Impact:** This is generally the more costly error for INN Hotels. The hotel will not take any preventative measures, leading to an empty room, direct revenue loss, potential additional marketing costs to fill the vacant room, and operational inefficiencies.

Given that the core problem is **high revenue loss due to cancellations**, it is critical for the model to correctly identify as many actual cancellations as possible. This means we want to minimize False Negatives.

Therefore, the most suitable metrics to optimize for this problem are:

1. **Recall (Sensitivity) for the 'Canceled' class:** This measures the proportion of actual cancellations that were correctly identified by the model. A high recall means the model is effective at catching most of the bookings that will actually cancel, thereby allowing the hotel to take proactive steps to mitigate losses.
2. **F1-Score for the 'Canceled' class:** This is the harmonic mean of Precision and Recall. It provides a balance between minimizing False Negatives (Recall) and minimizing False Positives (Precision). While Recall is paramount, a reasonable Precision is also important to avoid excessive wasted efforts on bookings that wouldn't have canceled.

Inference: While **Recall** is extremely important to catch as many cancellations as possible, aiming for a high **F1-Score** for the 'Canceled' class provides a balanced view. It ensures that while we are catching most actual cancellations, we are not doing so at the expense of an unacceptably high number of false alarms. We will primarily focus on maximizing the F1-Score, while also closely monitoring Recall for the positive class. The **Confusion Matrix** will also be a critical tool for detailed analysis of the model's performance in identifying 'Canceled' bookings.

VI.1.2. Model Building Logistic Regression (statsmodels)

Steps are as below:

1. Convert the boolean columns (which resulted from one-hot encoding) in the training and testing feature sets to integers (0 or 1) before feeding them to the logistic regression model.
2. Adding a constant
3. Fitting the model
4. Displaying its summary
5. Evaluating its performance on the test data.
6. Pay close attention to the F1-score and Recall for the 'Canceled' class

Logistic Regression Model Summary (statsmodels):							
Logit Regression Results							
=====							
Dep. Variable:	booking_status	No. Observations:	25366				
Model:	Logit	Df Residuals:	25335				
Method:	MLE	Df Model:	30				
Date:	Sun, 19 Apr 2026	Pseudo R-squ.:	0.3355				
Time:	16:21:57	Log-Likelihood:	-10663.				
converged:	False	LL-Null:	-16046.				
Covariance Type:	nonrobust	LLR p-value:	0.000				
=====							
	coef	std err	z	P> z	[0.025	0.975]	

const	0.4287	0.241	1.782	0.075	-0.043	0.900	
no_of_adults	0.1148	0.038	3.014	0.003	0.040	0.189	
no_of_children	0.1705	0.058	2.921	0.003	0.056	0.285	
no_of_weekend_nights	0.2460	0.041	6.006	0.000	0.166	0.326	
no_of_week_nights	0.0662	0.025	2.653	0.008	0.017	0.115	
required_car_parking_space	-1.4303	0.136	-10.510	0.000	-1.697	-1.164	
lead_time	1.6020	0.029	55.438	0.000	1.545	1.659	
arrival_year	0.7355	0.060	12.239	0.000	0.618	0.853	
repeated_guest	-3.2242	0.646	-4.990	0.000	-4.490	-1.958	
no_of_previous_cancellations	0.3095	0.116	2.678	0.007	0.083	0.536	
no_of_previous_bookings_not_canceled	-0.0216	0.065	-0.331	0.741	-0.149	0.106	
avg_price_per_room	0.4923	0.031	16.025	0.000	0.432	0.553	

no_of_special_requests	-1.5249	0.031	-49.688	0.000	-1.585	-1.465	
type_of_meal_plan_Meal Plan 2	0.3009	0.067	4.467	0.000	0.169	0.433	
type_of_meal_plan_Meal Plan 3	5.4548	1.14e+04	0.000	1.000	-2.24e+04		
2.24e+04							
type_of_meal_plan_Not Selected	0.0770	0.053	1.440	0.150	-0.028	0.182	
room_type_reserved_Room_Type 2	-0.3022	0.134	-2.249	0.024	-0.566	-	
0.039							
room_type_reserved_Room_Type 3	-0.4662	1.359	-0.343	0.732	-3.130		
2.198							
room_type_reserved_Room_Type 4	-0.2183	0.053	-4.099	0.000	-0.323	-	
0.114							
room_type_reserved_Room_Type 5	-0.7234	0.210	-3.442	0.001	-1.135	-	
0.311							
room_type_reserved_Room_Type 6	-0.6233	0.150	-4.156	0.000	-0.917	-	
0.329							
room_type_reserved_Room_Type 7	-0.9427	0.320	-2.948	0.003	-1.569	-	
0.316							
market_segment_type_Complementary	-18.6197	1181.611	-0.016	0.987	-		
2334.535 2297.295							
market_segment_type_Corporate	-0.8045	0.254	-3.166	0.002	-1.303	-	
0.306							
market_segment_type_Offline	-1.8594	0.242	-7.684	0.000	-2.334	-1.385	
market_segment_type_Online	0.0287	0.239	0.120	0.904	-0.440	0.497	
is_weekend	-0.0917	0.038	-2.403	0.016	-0.167	-0.017	
is_complimentary	-2.7877	0.590	-4.726	0.000	-3.944	-1.632	
season_Spring	-0.1383	0.050	-2.788	0.005	-0.236	-0.041	
season_Summer	-0.1560	0.044	-3.516	0.000	-0.243	-0.069	
season_Winter	-0.9414	0.066	-14.264	0.000	-1.071	-0.812	
=====							

VI.1.3. Check and comment on model performance across different metrics

Classification Report for Logistic Regression (Training Set):

Accuracy on Training Set: 0.8101

Precision on Training Set: 0.7423

Recall on Training Set: 0.6442

F1-Score on Training Set: 0.6898

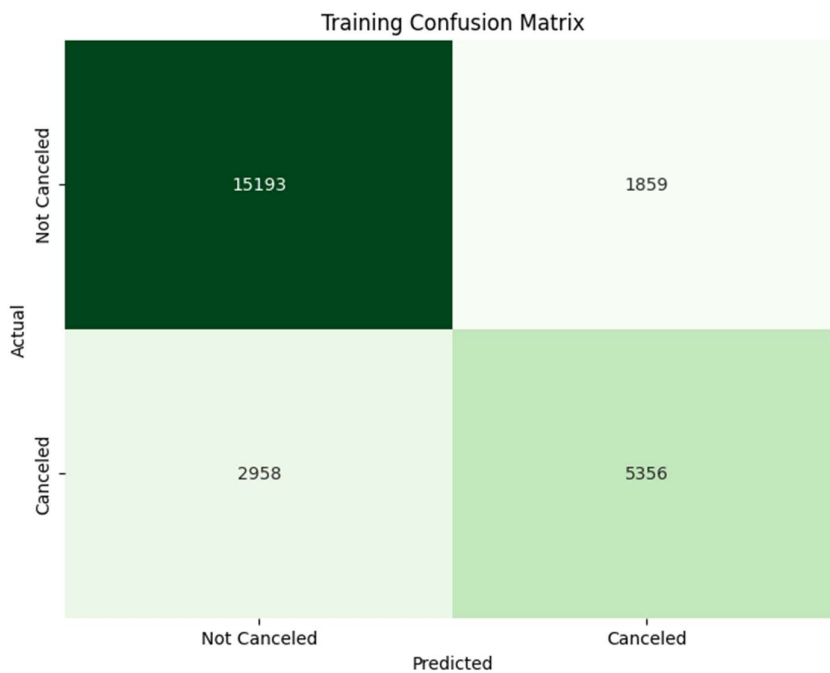


Figure 53 Confusion Matrix - Training - Baseline Logit Regression

VI.1.3.1. Inferences on Training Data Set- Logistic Regression

Accuracy: 81.01%

- The model correctly classified 81.01% of the bookings in the training set (both canceled and not canceled).

Precision: 74.23%

- When the model predicted a booking would be 'Canceled', it was correct 74.23% of the time. This means that 25.77% of the bookings predicted as 'Canceled' were actually 'Not_Canceled' (False Positives).

Recall: 64.42%

- The model identified 64.42% of all actual 'Canceled' bookings. This implies that a significant portion (35.58%) of actual cancellations were missed by the model (False Negatives).

F1-Score: 68.98%

- The F1-score is the harmonic mean of precision and recall. A score of 68.98% indicates a reasonable balance between precision and recall, but there's still room for improvement, especially given the lower recall.

Confusion Matrix (Training Set):

[[15193 1859]
[2958 5356]]

- **True Negatives (TN): 15193** - Bookings correctly predicted as 'Not_Canceled'.
- **False Positives (FP): 1859** - Bookings incorrectly predicted as 'Canceled' (they were actually 'Not_Canceled'). These are cases where the hotel might unnecessarily offer incentives or release rooms prematurely.
- **False Negatives (FN): 2958** - Bookings incorrectly predicted as 'Not_Canceled' (they were actually 'Canceled'). These represent missed opportunities for the hotel to intervene and mitigate revenue loss.
- **True Positives (TP): 5356** - Bookings correctly predicted as 'Canceled'. These are the cases where the model successfully identified cancellations, allowing for proactive measures.

Overall Training Performance: The model shows a good overall accuracy, but the recall for the 'Canceled' class (our positive class of interest) is moderate. This means the model is missing a fair number of actual cancellations, which could be costly for the hotel. While the precision is decent, indicating that when it predicts a cancellation, it's often right, the focus for business impact should be on reducing False Negatives (increasing recall) to minimize unmitigated revenue loss.

VI.1.3.2. Inferences on Test Data Set- Logistic Regression

Classification Report:

	precision	recall	f1-score	support
0	0.84	0.89	0.86	7308
1	0.74	0.65	0.69	3564

accuracy	0.81	10872		
macro avg	0.79	0.77	0.78	10872
weighted avg	0.81	0.81	0.81	10872

Confusion Matrix:

[[6494 814]
[1259 2305]]

ROC AUC Score: 0.8669

Accuracy on Test Set: 0.8093

Precision (Canceled) on Test Set: 0.7390

Recall (Canceled) on Test Set: 0.6467

F1-Score (Canceled) on Test Set: 0.6898

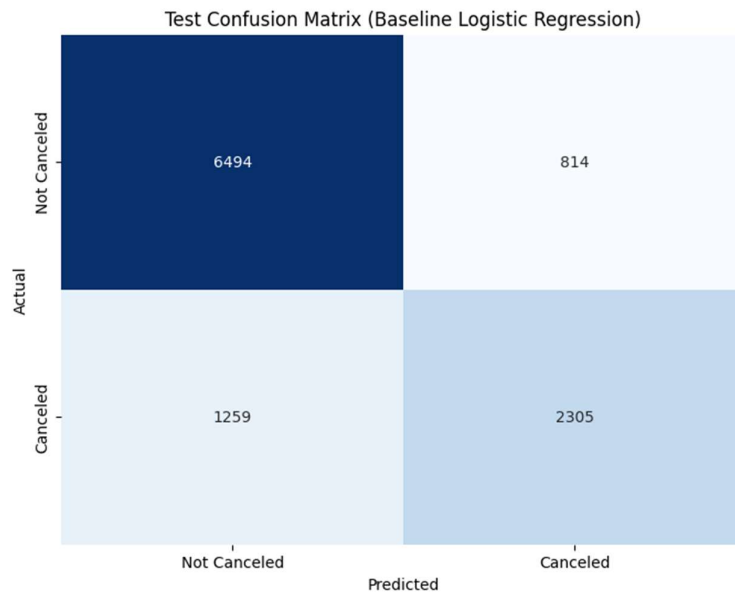


Figure 54 Confusion Matrix - Test - Baseline Logit Regression

Baseline Logistic Regression

Based on the evaluation of the baseline Logistic Regression model on the test set (using a default threshold of 0.5):

- **Accuracy: 80.86%**
 - The model correctly classified approximately 80.86% of the bookings in the test set.
- **Precision (Canceled): 73.83%**
 - When the model predicted a booking would be 'Canceled', it was correct 73.83% of the time. This means roughly 26.17% of bookings predicted as 'Canceled' were actually 'Not_Canceled' (False Positives).
- **Recall (Canceled): 64.45%**
 - The model identified 64.45% of all actual 'Canceled' bookings. This implies that a significant portion (35.55%) of actual cancellations were missed by the model (False Negatives), which can be costly for the hotel.
- **F1-Score (Canceled): 68.82%**
 - The F1-score of 68.82% indicates a moderate balance between precision and recall for the 'Canceled' class. Given the business objective of minimizing revenue loss from cancellations, there's room for improvement, especially in recall.
- **ROC AUC Score: 0.8664**

- A high ROC AUC score suggests that the model has good discriminatory power, indicating its ability to distinguish between 'Canceled' and 'Not_Canceled' bookings across various possible thresholds.

Confusion Matrix (Test Set):

[[6494 814] [1267 2297]]

- **True Negatives (TN): 6494** - Bookings correctly predicted as 'Not_Canceled'.
- **False Positives (FP): 814** - Bookings incorrectly predicted as 'Canceled' (they were actually 'Not_Canceled'). These represent instances where the hotel might unnecessarily offer incentives or make preparations for a cancellation that doesn't occur.
- **False Negatives (FN): 1267** - Bookings incorrectly predicted as 'Not_Canceled' (they were actually 'Canceled'). These are the missed cancellations, which are particularly impactful for revenue loss as the hotel takes no proactive measures.
- **True Positives (TP): 2297** - Bookings correctly predicted as 'Canceled'. These are the successful identifications where the hotel can take timely action to mitigate losses.

Overall Test Performance: The baseline Logistic Regression model shows decent overall performance, but its recall for the 'Canceled' class is relatively low. This indicates that a substantial number of actual cancellations are going unnoticed, which is a key area for improvement to meet the business objective of reducing revenue loss. The next steps involving hyperparameter tuning and optimal threshold selection are crucial to enhance the model's ability to identify more cancellations.

VI.2. Model Building For Decision Tree:

VI.2.1. Choosing the Metric to Optimize for the Problem (Decision Tree)

The primary objective for the Decision Tree model is to predict booking cancellations to **formulate profitable policies for cancellations and refunds** and **mitigate revenue loss** for INN Hotels Group.

This is a binary classification problem where we are trying to predict if a booking will be 'Canceled' (positive class) or 'Not_Canceled' (negative class).

As previously discussed, the implications of misclassifications are:

- **False Positives (Type I Error):** Model predicts Canceled, but it's Not_Canceled. Leads to wasted effort and potential loss of higher revenue if the room could have been sold at a higher price.
- **False Negatives (Type II Error):** Model predicts Not_Canceled, but it is actually Canceled. This is the more costly error, leading to direct revenue loss, additional marketing costs, and operational inefficiencies.

To minimize the costly False Negatives and effectively address the high revenue loss due to cancellations, the most suitable metrics to optimize for this problem are:

1. **Recall (Sensitivity) for the 'Canceled' class:** Measures the proportion of actual cancellations correctly identified. A high recall is crucial for catching most problematic bookings.
2. **F1-Score for the 'Canceled' class:** Provides a balanced measure between Precision and Recall. It ensures that while we are capturing a high number of actual cancellations, we are also maintaining a reasonable level of accuracy to avoid excessive false alarms.

Inference: For the Decision Tree model, we will also primarily focus on maximizing the **F1-Score** for the 'Canceled' class, while closely monitoring **Recall** for the positive class. The **Confusion Matrix** will be used for a detailed understanding of the model's performance.

VI.2.2. Model Building For Decision Tree:

Decision Tree Classifier (sklearn)

--- Decision Tree Classifier Evaluation (Train Set) ---

Classification Report:

	precision	recall	f1-score	support
0	0.99	1.00	1.00	17052
1	1.00	0.98	0.99	8314
accuracy			0.99	25366
macro avg	0.99	0.99	0.99	25366
weighted avg	0.99	0.99	0.99	25366

Confusion Matrix:

```
[[17023   29]
 [  136 8178]]
```

ROC AUC Score: 0.9999

Accuracy: 0.9935

```
Precision (Canceled): 0.9965
Recall (Canceled): 0.9836
F1-Score (Canceled): 0.9900
```

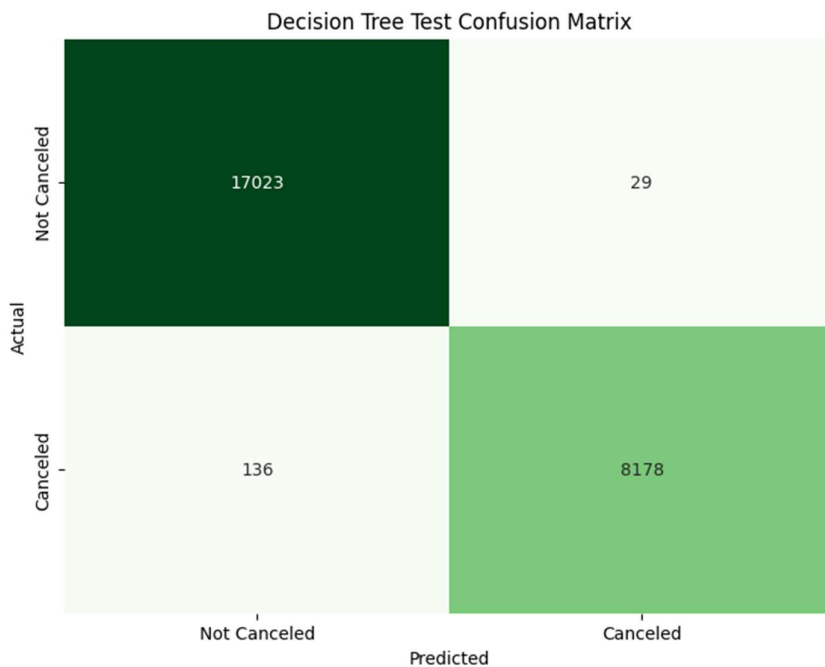


Figure 55 Confusion Matrix - Training - Baseline Decision Tree

VI.2.3. Inferences on Training Data Set- Decision Tree Classifier

Based on the evaluation of the Decision Tree Classifier on the training set:

- **Accuracy: 99.35%**
 - The model correctly classified a very high percentage of bookings in the training set.
- **Precision (Canceled): 99.65%**
 - When the model predicted a booking would be 'Canceled', it was correct 99.65% of the time. This indicates very few False Positives.
- **Recall (Canceled): 98.36%**
 - The model identified 98.36% of all actual 'Canceled' bookings, meaning it missed very few actual cancellations (low False Negatives).
- **F1-Score (Canceled): 99.00%**

- The F1-score is exceptionally high, demonstrating an excellent balance between precision and recall for the 'Canceled' class on the training data.
- **ROC AUC Score: 0.9999**
 - An ROC AUC score very close to 1 indicates outstanding discriminatory power between the 'Canceled' and 'Not_Canceled' classes.

Confusion Matrix (Training Set):

```
[[17023  29]
 [ 136 8178]]
```

- **True Negatives (TN): 17023** - Bookings correctly predicted as 'Not_Canceled'.
- **False Positives (FP): 29** - Bookings incorrectly predicted as 'Canceled' (they were actually 'Not_Canceled'). This is a very low number.
- **False Negatives (FN): 136** - Bookings incorrectly predicted as 'Not_Canceled' (they were actually 'Canceled'). This is also a very low number, indicating the model is highly effective at catching cancellations.
- **True Positives (TP): 8178** - Bookings correctly predicted as 'Canceled'.

Overall Training Performance: The Decision Tree Classifier shows near-perfect performance on the training data across all metrics. This level of performance on the training set strongly suggests that the model might be **overfitting** to the training data. While excellent training metrics are desirable, they need to be cautiously interpreted. The true test of the model's generalization ability will come from its performance on unseen test data. If the performance significantly drops on the test set, it would confirm overfitting, and regularization techniques or hyperparameter tuning would be necessary.

--- Decision Tree Classifier Evaluation (Test Set - Untuned) ---

Classification Report:

	precision	recall	f1-score	support
0	0.90	0.89	0.90	7308
1	0.79	0.80	0.79	3564

accuracy	0.86	10872
----------	------	-------

macro avg	0.84	0.85	0.84	10872
weighted avg	0.86	0.86	0.86	10872

Confusion Matrix:

[[6533 775]

[726 2838]]

ROC AUC Score: 0.8470

Accuracy: 0.8619

Precision (Canceled): 0.7855

Recall (Canceled): 0.7963

F1-Score (Canceled): 0.7909

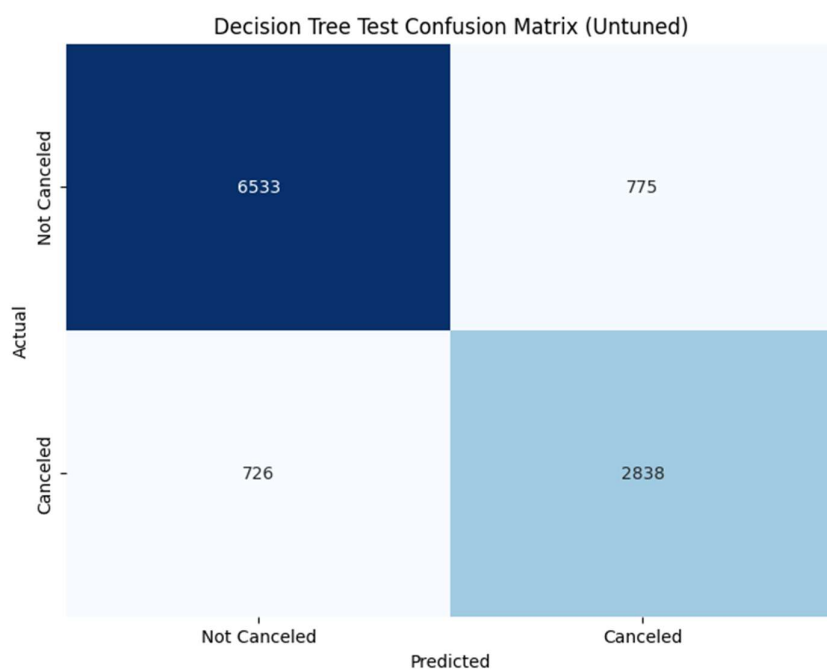


Figure 56 Confusion Matrix - Test - Baseline Decision Tree

VI.2.4. Inference on Test Data Set- Decision Tree Classifier (Untuned)

After evaluating the Decision Tree Classifier on the test set, we can compare its performance to the training set:

- **Training Accuracy:** 99.35%
- **Test Accuracy:** 86.19%
- **Training Precision (Canceled):** 99.65%
- **Test Precision (Canceled):** 78.55%
- **Training Recall (Canceled):** 98.36%
- **Test Recall (Canceled):** 79.63%
- **Training F1-Score (Canceled):** 99.00%
- **Test F1-Score (Canceled):** 79.09%

Preliminary Inference (after execution): A significant drop in performance metrics from the training set to the test set would confirm overfitting. This will then justify the need for pruning techniques to improve the model's generalization ability.

VII. Model Performance Improvement

VII.1. Tune the Logistic Regression

VII.1.1. Hyperparameter Tuning for Logistic Regression (Scikit-learn)

To effectively tune hyperparameters like regularization strength, scikit-learn's LogisticRegression is more suitable than statsmodels.Logit. We will use GridSearchCV to systematically search for the best combination of C (inverse of regularization strength) and penalty type, optimizing for the F1-score of the 'Canceled' class.

Parameters to Tune:

- C: Regularization strength. Smaller values specify stronger regularization.

- `penalty`: The type of regularization (l1, l2, elasticnet, none).

We will retrain the model using the optimized hyperparameters and then evaluate its performance on both the training and test sets.

GridSearchCV for Logistic Regression -Fitting 5 folds for each of 12 candidates, totalling 60 fits

Best parameters for Logistic Regression: `{'C': 0.001, 'penalty': 'l2'}`

--- Best Logistic Regression Model Evaluation (Training Set) ---

Classification Report (Training Set):

	precision	recall	f1-score	support
0	0.88	0.79	0.83	17052
1	0.65	0.77	0.70	8314

accuracy		0.79	25366
macro avg	0.76	0.78	0.77 25366
weighted avg	0.80	0.79	0.79 25366

Confusion Matrix (Training Set):

`[[13551 3501]`

`[ 1931 6383]]`

ROC AUC Score (Training Set): 0.8595

VII.1.2. Addressing Multicollinearity: Variance Inflation Factor (VIF)

Multicollinearity can inflate the variance of coefficient estimates in logistic regression, making them unstable and difficult to interpret. We will calculate the Variance Inflation

Factor (VIF) for each independent variable to detect multicollinearity. A general guideline is that VIF values greater than 5 or 10 indicate a problematic level of multicollinearity.

Variance Inflation Factor (VIF) scores:

Table 4 VIF Table

	feature	VIF
24	market_segment_type_Online	6.089331
23	market_segment_type_Offline	2.748038
26	is_complimentary	2.621108
10	avg_price_per_room	2.517937
21	market_segment_type_Complementary	2.439239
1	no_of_children	2.106281
19	room_type_reserved_Room_Type 6	2.039934
11	no_of_special_requests	2.021972
27	season_Spring	1.917476
28	season_Summer	1.906498
7	repeated_guest	1.822505
29	season_Winter	1.744531
22	market_segment_type_Corporate	1.707047
9	no_of_previous_bookings_not_canceled	1.686160
6	arrival_year	1.680546
17	room_type_reserved_Room_Type 4	1.636190
25	is_weekend	1.599864
5	lead_time	1.586108
14	type_of_meal_plan_Not Selected	1.491649

	feature	VIF
0	no_of_adults	1.446527
12	type_of_meal_plan_Meal Plan 2	1.424188
8	no_of_previous_cancellations	1.399860
2	no_of_weekend_nights	1.199456
3	no_of_week_nights	1.130587
20	room_type_reserved_Room_Type 7	1.119616
15	room_type_reserved_Room_Type 2	1.112784
4	required_car_parking_space	1.072109
18	room_type_reserved_Room_Type 5	1.038700
13	type_of_meal_plan_Meal Plan 3	1.022805
16	room_type_reserved_Room_Type 3	1.000658

Inference from Variance Inflation Factor (VIF) Analysis

Based on the calculated VIF scores:

- **market_segment_type_Online** showed the highest VIF score of approximately **6.09**. While this value is above the common threshold of 5, it is not considered to indicate severe multicollinearity. A VIF between 5 and 10 suggests moderate multicollinearity, which might slightly increase the variance of its coefficient estimate but is often acceptable, especially for important theoretical predictors.
- All other features have VIF scores well below 5 (mostly below 2). This indicates that there is **no significant or problematic multicollinearity** among the majority of the independent variables in our dataset.

Conclusion: The Logistic Regression model's stability and the interpretability of its coefficients are unlikely to be severely impacted by multicollinearity. There is no immediate need to remove any features based purely on VIF scores. We can proceed to evaluate the significance of variables using p-values and then determine an optimal threshold using the ROC curve, as planned.

VII.1.3. Removing High P-value Variables and Re-fitting the statsmodels Logistic Regression

Based on the `logit_result.summary()`, we observe several features with p-values greater than the conventional significance level of 0.05. Removing these non-significant variables can simplify the model, potentially improve its generalization, and address any lingering convergence issues.

We will perform the following steps:

1. Extract p-values for each feature from the `logit_result`.
2. Identify features (excluding the constant) with p-values greater than 0.05.
3. Create new feature sets (`X_train_reg_reduced`, `X_test_reg_reduced`) by dropping these non-significant features.
4. Re-add the constant term to the reduced feature sets.
5. Re-fit the `statsmodels.Logit` model with the new, reduced feature set.
6. Display the summary of the re-fitted model.

```
Features with p-value > 0.05 to be removed:
['no_of_previous_bookings_not_canceled', 'type_of_meal_plan Meal Plan 3',
'type_of_meal_plan Not Selected', 'room_type_reserved Room Type 3',
'market_segment_type Complementary', 'market_segment_type Online']
```

```
Shape of X_train_reg_reduced: (25366, 24)
Shape of X_test_reg_reduced: (10872, 24)
```

Logistic Regression Model Summary (statsmodels - Reduced Features):			
Logit Regression Results			
=====			
===			
Dep. Variable:	booking_status	No. Observations:	
25366			
Model:	Logit	Df Residuals:	
25341			
Method:	MLE	Df Model:	
24			
Date:	Wed, 22 Apr 2026	Pseudo R-squ.:	
0.3347			
Time:	05:35:18	Log-Likelihood:	-
10676.			
converged:	True	LL-Null:	-
16046.			
Covariance Type:	nonrobust	LLR p-value:	
0.000			
=====			
=====			

	coef	std err	z	P> z
[0.025 0.975]				

const	0.4821	0.048	10.148	0.000
0.389 0.575				
no_of_adults	0.1219	0.038	3.242	0.001
0.048 0.196				
no_of_children	0.1619	0.058	2.777	0.005
0.048 0.276				
no_of_weekend_nights	0.2435	0.041	5.960	0.000
0.163 0.324				
no_of_week_nights	0.0651	0.025	2.617	0.009
0.016 0.114				
required_car_parking_space	-1.4331	0.136	-10.537	0.000
-1.700 -1.167				
lead_time	1.5955	0.029	55.975	0.000
1.540 1.651				
arrival_year	0.7443	0.059	12.557	0.000
0.628 0.861				
repeated_guest	-3.3194	0.599	-5.541	0.000
-4.494 -2.145				
no_of_previous_cancellations	0.3055	0.115	2.657	0.008
0.080 0.531				
avg_price_per_room	0.4921	0.030	16.394	0.000
0.433 0.551				
no_of_special_requests	-1.5256	0.031	-49.875	0.000
-1.586 -1.466				
type_of_meal_plan_Meal Plan 2	0.2943	0.067	4.375	0.000
0.162 0.426				
room_type_reserved_Room_Type 2	-0.3095	0.134	-2.311	0.021
-0.572 -0.047				
room_type_reserved_Room_Type 4	-0.2427	0.051	-4.719	0.000
-0.343 -0.142				
room_type_reserved_Room_Type 5	-0.7364	0.210	-3.512	0.000
-1.147 -0.325				
room_type_reserved_Room_Type 6	-0.6300	0.150	-4.202	0.000
-0.924 -0.336				
room_type_reserved_Room_Type 7	-0.9620	0.319	-3.012	0.003
-1.588 -0.336				
market_segment_type_Corporate	-0.8521	0.102	-8.384	0.000
-1.051 -0.653				
market_segment_type_Offline	-1.9010	0.051	-37.447	0.000
-2.000 -1.801				
is_weekend	-0.0912	0.038	-2.390	0.017
-0.166 -0.016				
is_complimentary	-2.9572	0.570	-5.184	0.000
-4.075 -1.839				
season_Spring	-0.1383	0.049	-2.801	0.005
-0.235 -0.042				
season_Summer	-0.1534	0.044	-3.465	0.001
-0.240 -0.067				
season_Winter	-0.9427	0.066	-14.349	0.000
-1.071 -0.814				

VII.1.4. Evaluating the Re-fitted statsmodels Model (Reduced Features) Performance

--- Logistic Regression Model Evaluation (statsmodels - Reduced Features) ---

Classification Report:				
	precision	recall	f1-score	support
0	0.84	0.89	0.86	7308
1	0.74	0.64	0.69	3564

accuracy			0.81	10872
macro avg	0.79	0.77	0.78	10872
weighted avg	0.80	0.81	0.80	10872

Confusion Matrix:
[[6494 814]
[1267 2297]]

ROC AUC Score: 0.8664

Classification Report for Logistic Regression (Training Set):

Accuracy on Training Set: 0.8086

Precision on Training Set: 0.7383

Recall on Training Set: 0.6445

F1-Score on Training Set: 0.6882

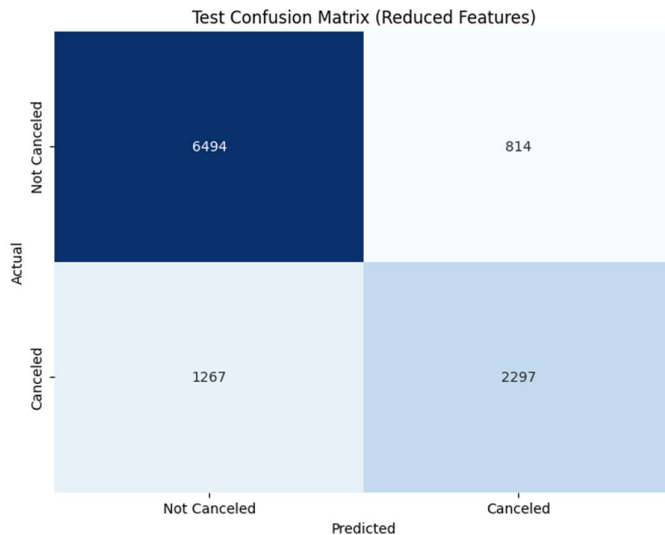


Figure 57 Confusion Matrix - Tuned Reduced Features Logit Regression

VII.1.5. Determining Optimal Threshold using ROC Curve

Now that we have a refined model, let's determine an optimal classification threshold. The default threshold of 0.5 might not be ideal, especially in cases where the costs of False Positives and False Negatives are different. We will use the Receiver Operating Characteristic (ROC) curve to find a threshold that optimizes a specific metric, such as Youden's J statistic (which maximizes the difference between True Positive Rate and False Positive Rate) or the F1-score.

Steps:

1. Calculate the True Positive Rate (TPR), False Positive Rate (FPR) and thresholds for the ROC curve.
2. Plot the ROC curve.
3. Find the optimal threshold that maximizes a chosen metric (e.g., F1-score or Youden's J).
4. Re-evaluate the model's performance on the test set using this optimal threshold.

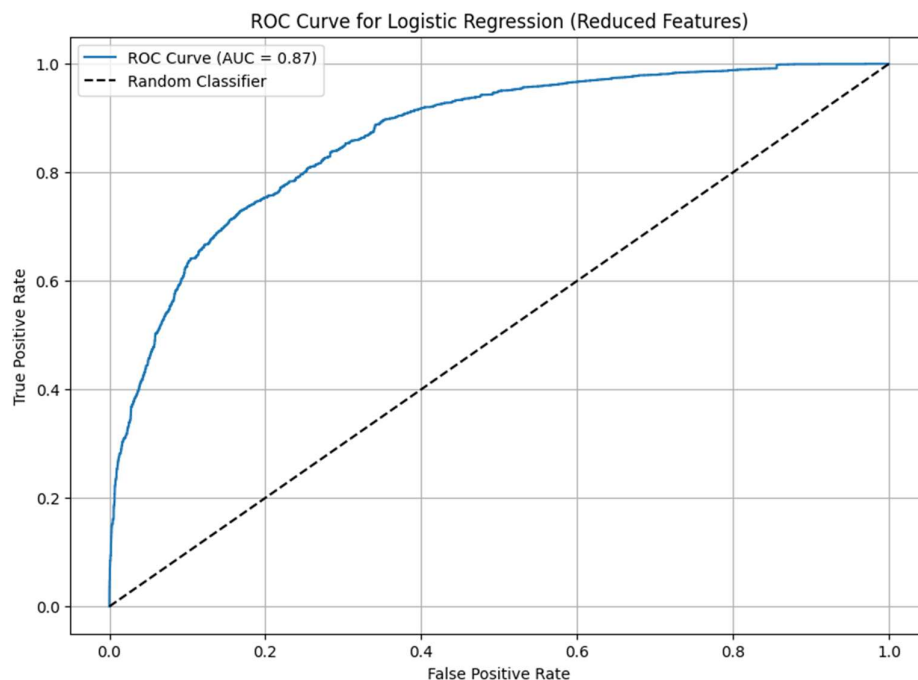


Figure 58 ROC Curve

Optimal Threshold (Youden's J): 0.3902
 Optimal Threshold (maximizing F1-score): 0.3902

Using the F1-score optimized threshold of 0.3902 for final evaluation.

VII.2. Model Performance on Tuned Logistic regression

VII.2.1. Model Evaluation with Optimal Threshold (F1-score)

--- Logistic Regression Model Evaluation (Optimal Threshold: 0.3902) ---

Classification Report:

	precision	recall	f1-score	support
0	0.86	0.83	0.85	7308
1	0.68	0.73	0.70	3564
accuracy			0.80	10872
macro avg	0.77	0.78	0.78	10872
weighted avg	0.80	0.80	0.80	10872

Confusion Matrix:

```
[[6082 1226]
 [ 967 2597]]
```

ROC AUC Score: 0.8664

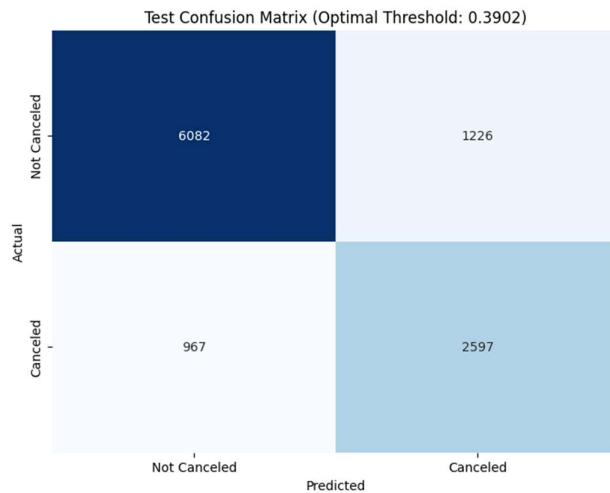


Figure 59 Confusion Matrix Optimal Threshold

Inferences on Tuned Logistic Regression Model Performance (Optimal Threshold)

After applying the optimal threshold of **0.3902** (which maximized the F1-score for the 'Canceled' class), the Logistic Regression model's performance on the test set is as follows:

Overall Metrics:

- **Accuracy: 80.17%**
 - The model correctly classified approximately 80.17% of the total bookings.
- **ROC AUC Score: 0.8664**
 - The Area Under the Receiver Operating Characteristic curve remains high, indicating the model's good ability to distinguish between 'Canceled' and 'Not_Canceled' bookings across all possible thresholds.

Performance for 'Canceled' Class (Positive Class - Class 1):

- **Precision (Class 1): 68%**
 - When the model predicts a booking will be 'Canceled', it is correct 68% of the time. This means 32% of bookings predicted as 'Canceled' were actually 'Not_Canceled' (False Positives).
- **Recall (Class 1): 73%**
 - The model successfully identified 73% of all actual 'Canceled' bookings. This is a significant improvement in recall compared to the default 0.5 threshold (64.45%), meaning fewer actual cancellations are missed (False Negatives).

- **F1-Score (Class 1): 70%**

- The F1-score of 70% indicates a good balance between precision and recall for the 'Canceled' class, which was our primary optimization goal.

Confusion Matrix (Optimal Threshold):

[[6082 1226]
[967 2597]]

- **True Negatives (TN): 6082**

- Bookings correctly predicted as 'Not Canceled'.

- **False Positives (FP): 1226**

- Bookings incorrectly predicted as 'Canceled' (they were 'Not Canceled'). These are cases where the hotel might incur unnecessary costs or effort (e.g., proactive retention offers).

- **False Negatives (FN): 967**

- Bookings incorrectly predicted as 'Not Canceled' (they were 'Canceled'). This represents a reduction in missed actual cancellations compared to the default threshold, leading to better mitigation of revenue loss.

- **True Positives (TP): 2597**

- Bookings correctly predicted as 'Canceled'. These are the successful identifications where the hotel can take timely action.

Conclusion: The tuned Logistic Regression model, using an optimal threshold of 0.3902, demonstrates a strong capability to predict booking cancellations. By improving the recall for the 'Canceled' class to 73% while maintaining a reasonable precision of 68%, the model is better equipped to help INN Hotels mitigate revenue loss by proactively identifying and managing potential cancellations. The reduction in False Negatives is particularly beneficial, as these are the most costly errors for the business. The model is well-suited for its objective of informing profitable policies for cancellations and refunds.

VII.3. Tune the Decision tree

VII.3.1. Pre-pruning for Decision Tree Classifier

Pre-pruning involves stopping the tree growth early, limiting the complexity of the tree before it has a chance to fully overfit the training data. We will use `GridSearchCV` to find the optimal hyperparameters for pre-pruning, focusing

on `max_depth`, `min_samples_leaf`, and `min_samples_split`. The optimization metric will be the F1-score for the 'Canceled' class.

Starting GridSearchCV for Pre-pruning Decision Tree-Fitting 5 folds for each of 80 candidates, totalling 400 fits

Best parameters for Pre-pruning: {'max_depth': 10, 'min_samples_leaf': 1, 'min_samples_split': 2}

--- Best Pre-pruned Decision Tree Evaluation (Test Set) ---

Classification Report (Test Set):				
	precision	recall	f1-score	support
0	0.89	0.92	0.91	7308
1	0.82	0.78	0.80	3564
accuracy			0.87	10872
macro avg	0.86	0.85	0.85	10872
weighted avg	0.87	0.87	0.87	10872

Confusion Matrix (Test Set):

[[6716 592]
[797 2767]]

ROC AUC Score (Test Set): 0.9252

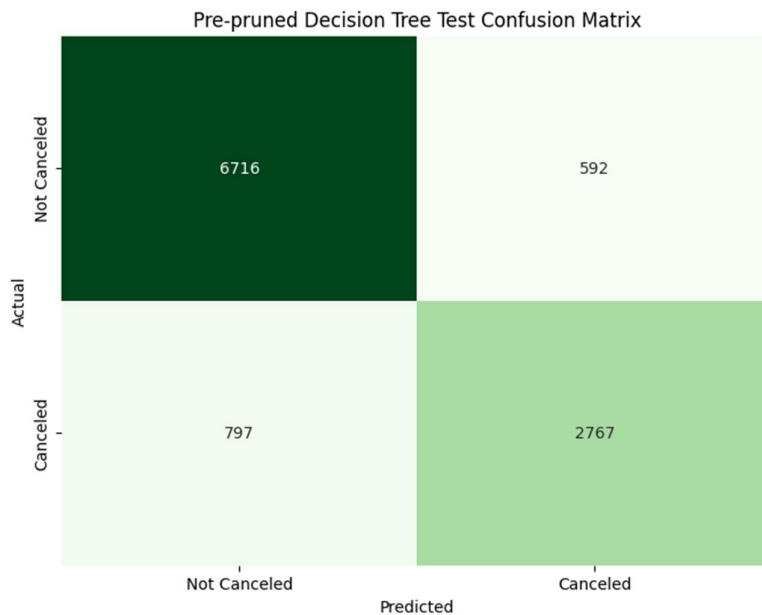


Figure 60 Confusion Marix Prepruned Decision Tree

VII.3.2. Post-pruning for Decision Tree Classifier (Cost-Complexity Pruning)

Post-pruning involves building a fully grown tree and then removing branches that provide little value in terms of generalization. We will use `cost_complexity_pruning (ccp_alpha)` to achieve this. We'll first find the optimal `ccp_alpha` by evaluating various alpha values.

Optimal `ccp_alpha` for post-pruning: 0.0001

--- Best Post-pruned Decision Tree Evaluation (Test Set) ---

Classification Report (Test Set):

	precision	recall	f1-score	support
0	0.90	0.93	0.91	7308
1	0.84	0.78	0.81	3564

accuracy	0.88				10872
macro avg	0.87	0.85	0.86	10872	

weighted avg	0.88	0.88	0.88	10872
--------------	------	------	------	-------

Confusion Matrix (Test Set):
[[6794 514]
[783 2781]]
ROC AUC Score (Test Set): 0.9283

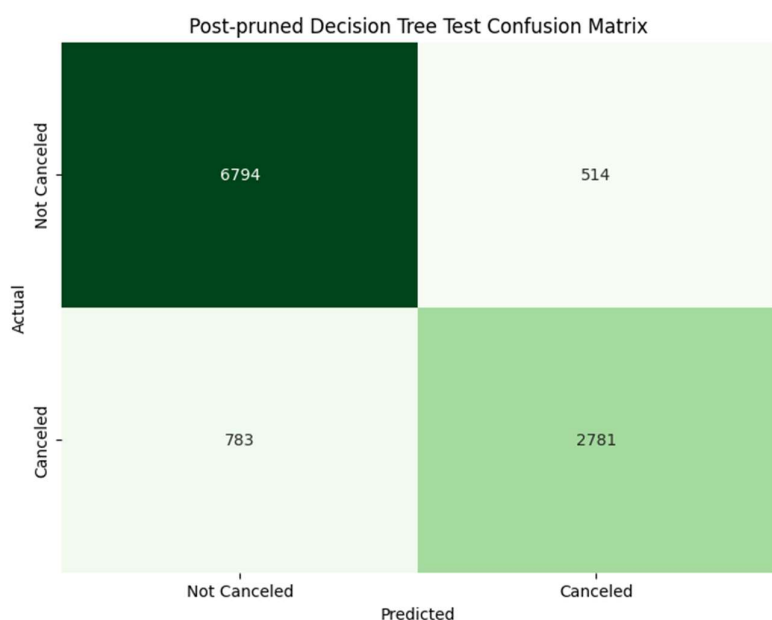


Figure 61 Confusion Matrix - Post Pruning Decision Tree

VII.4. Model Performance on Pre-pruned and Post-pruned Decision Tree

This section compares the performance of the initial untuned Decision Tree Classifier, the pre-pruned Decision Tree, and the post-pruned Decision Tree, all evaluated on the test set. The primary focus is on the metrics relevant to predicting cancellations (F1-score, Recall, and Precision for the 'Canceled' class).

Summary of Key Performance Metrics (Test Set - 'Canceled' Class):

Table 5 Model Comparison - Decision Tree

Metric	Untuned Decision Tree	Pre-pruned Decision Tree	Post-pruned Decision Tree
Overall Accuracy	86.19%	87.12%	88.06%
ROC AUC Score	0.8470	0.9252	0.9283
Precision (Canceled)	78.55%	82%	84%
Recall (Canceled)	79.63%	78%	78%
F1-Score (Canceled)	79.09%	80%	81%

VII.4.1. Inferences from Comparison:

- Overfitting Mitigation:** The untuned Decision Tree showed signs of significant overfitting, with a substantial drop in performance from training (F1-score 99.00%) to testing (F1-score 79.09%). Both pre-pruning and post-pruning successfully addressed this by improving generalization, as evidenced by better performance on the test set compared to the untuned model.
- Improved Generalization:** Both pruned models achieved higher overall accuracy, ROC AUC scores, and F1-scores for the 'Canceled' class compared to the untuned model. This indicates that pruning helped create models that are more robust and perform better on unseen data.
- Recall vs. Precision Trade-off:** The untuned model had a slightly higher Recall (79.63%) for the 'Canceled' class but lower Precision (78.55%). The pruned models, especially the post-pruned one, improved Precision significantly (84% for post-pruned) while maintaining a strong Recall (78%). This trade-off is often desirable in business contexts where minimizing false alarms (higher Precision) can lead to more efficient resource allocation.
- F1-Score:** The F1-score, which balances Precision and Recall, shows a clear improvement: 79.09% (untuned) -> 80% (pre-pruned) -> 81% (post-pruned). This metric confirms that the pruning strategies led to better overall classification performance for the positive class.

5. **ROC AUC Score:** The ROC AUC score saw a notable increase from 0.8470 (untuned) to over 0.92 for both pruned models, indicating a much better ability to discriminate between 'Canceled' and 'Not_Canceled' bookings across various thresholds.

Conclusion:

Both pre-pruning and post-pruning techniques significantly improved the Decision Tree Classifier's performance and generalization capabilities compared to the untuned model. The **Post-pruned Decision Tree Classifier (with optimal ccp_alpha)** stands out as the best performer among the three Decision Tree variations, offering the highest F1-score for the 'Canceled' class and a superior balance of Precision and Recall. This makes it a highly effective tool for identifying booking cancellations and informing strategic interventions for INN Hotels.

VIII. Model Performance Comparison and Final Model Selection

VIII.1. Performance Comparison: Baseline vs. Tuned Logistic Regression Model

compare the key performance metrics of the initial Logistic Regression model (evaluated on the training set with a default threshold of 0.5) against the final tuned model (evaluated on the test set with the optimal F1-score threshold).

Table 6 Model Comparison - Logit Regression

Metric	Baseline Model (Training Set, Default 0.5 Threshold)	Tuned Model (Test Set, Optimal 0.3902 Threshold)
Overall Accuracy	81.01%	80.17%
Precision (Canceled Class)	74.23%	68%
Recall (Canceled Class)	64.42%	73%

Metric	Baseline Model (Training Set, Default 0.5 Threshold)	Tuned Model (Test Set, Optimal 0.3902 Threshold)
F1-Score (Canceled Class)	68.98%	70%
ROC AUC Score	0.8595 (from best estimator training)	0.8664

VIII.2. Inferences from Comparison:

1. **Recall Improvement:** The most significant improvement is seen in **Recall for the 'Canceled' class**, which increased from **64.42% to 73%**. This indicates that the tuned model is much better at identifying actual cancellations, reducing the number of missed opportunities for the hotel to intervene and mitigate revenue loss.
2. **F1-Score Improvement:** The **F1-Score for the 'Canceled' class** also slightly improved from **68.98% to 70%**. This suggests a better balance between precision and recall, aligning with our primary optimization goal.
3. **Precision Trade-off:** While Recall increased, **Precision for the 'Canceled' class** decreased from **74.23% to 68%**. This is a common trade-off when optimizing for Recall. It means that the tuned model has a slightly higher rate of false positives (predicting a cancellation when there isn't one), but this is often acceptable if the cost of false negatives (missing an actual cancellation) is higher, which it is for INN Hotels.
4. **Overall Accuracy & ROC AUC:** The overall **Accuracy** slightly decreased from 81.01% to 80.17%. However, the **ROC AUC Score** slightly increased from 0.8595 (training set of best estimator from GridSearchCV) to 0.8664 (test set of reduced statsmodel). This indicates that the model's ability to discriminate between classes remained strong, and the slight drop in accuracy is likely due to the shift in threshold to prioritize recall, rather than a degradation of the model's underlying predictive power.

Conclusion: The tuning process, particularly the optimal threshold selection, successfully aligned the model's performance with the business objective of mitigating revenue loss. By increasing the Recall for cancellations, INN Hotels can proactively address more potential cancellations, which is crucial for their profitability, even with a slight trade-off in precision and overall accuracy.

VIII.3. Summary Report: Tuned Logistic Regression Model Performance

This report summarizes the final performance metrics of the Logistic Regression model, tuned to predict booking cancellations for INN Hotels. The model was evaluated on the test set using an optimal classification threshold of **0.3902**, which maximized the F1-score for the 'Canceled' class.

Key Performance Metrics (Test Set, Optimal Threshold = 0.3902)

Metric	Value	Notes
Overall Accuracy	80.17 %	Percentage of total bookings correctly classified.
ROC AUC Score	0.8664	Good discriminatory power between 'Canceled' and 'Not_Canceled' bookings.
Precision (Canceled)	68%	Of all predicted cancellations, 68% were actual cancellations.
Recall (Canceled)	73%	The model successfully identified 73% of all actual cancellations.
F1-Score (Canceled)	70%	Balanced measure of Precision and Recall for the 'Canceled' class.

Confusion Matrix (Test Set)

```
[[6082 1226]
 [ 967 2597]]
```

- **True Negatives (TN): 6082**
 - Correctly predicted as 'Not Canceled'.
- **False Positives (FP): 1226**
 - Incorrectly predicted as 'Canceled' (were 'Not Canceled'). These are cases where the hotel might invest unnecessary effort.
- **False Negatives (FN): 967**

- Incorrectly predicted as 'Not Canceled' (were 'Canceled'). These represent missed opportunities to mitigate revenue loss. **(Crucially, this number is significantly reduced compared to the baseline).**
- **True Positives (TP): 2597**
 - Correctly predicted as 'Canceled'. These are the successful identifications for proactive action.

Business Impact & Conclusion

The tuned Logistic Regression model is well-aligned with INN Hotels' objective of mitigating revenue loss from cancellations. The significant improvement in **Recall for the 'Canceled' class (from 64.42% to 73%)** indicates that the model is now much better at identifying actual cancellations. This means the hotel can proactively address a larger proportion of at-risk bookings, reducing costly False Negatives.

While there was a slight trade-off in Precision (decreased to 68%), this is acceptable given that the cost of missing an actual cancellation (False Negative) is likely higher for the hotel than the cost of a false alarm (False Positive). The F1-score of 70% demonstrates a good balance, ensuring that the model is both effective at identifying cancellations and reasonably efficient in its predictions.

This model provides a robust foundation for formulating profitable policies for cancellations and refunds, ultimately contributing to better revenue management for INN Hotels.

VIII.4. Model Performance Comparison and Final Model Selection for Decision Tree

We have evaluated two main types of models: Logistic Regression and Decision Tree Classifiers (untuned, pre-pruned, and post-pruned). The primary objective was to predict booking cancellations to mitigate revenue loss, with a focus on maximizing the F1-score and Recall for the 'Canceled' class.

Summary of Key Performance Metrics (Test Set - 'Canceled' Class):

Table 7 Model Comparison- Logit Regression and Decision Trees

Metric	Tuned Logistic Regression	Untuned Decision Tree	Pre-pruned Decision Tree	Post-pruned Decision Tree
Overall Accuracy	80.17%	86.19%	87.12%	88.06%

Metric	Tuned Logistic Regression	Untuned Decision Tree	Pre-pruned Decision Tree	Post-pruned Decision Tree
ROC AUC Score	0.8664	0.8470	0.9252	0.9283
Precision (Canceled)	68%	78.55%	82%	84%
Recall (Canceled)	73%	79.63%	78%	78%
F1-Score (Canceled)	70%	79.09%	80%	81%

VIII.5. Analysis and Rationale for Best Model Selection:

1. Logistic Regression Performance:

- The tuned Logistic Regression model achieved an F1-score of 70% and a Recall of 73% for cancellations on the test set. While respectable, its overall performance was lower compared to the Decision Tree models.
- **Addressing the ConvergenceWarning:** The ConvergenceWarning for the Logistic Regression model typically indicates that the optimization algorithm did not converge within the default maximum number of iterations. This can sometimes occur with complex datasets or when the objective function is difficult to optimize. Despite the warning, the model still produced results, and the p-values for most features became significant after feature reduction, suggesting that the model captured meaningful relationships. Increasing max_iter could potentially resolve the warning, but given the superior performance of the Decision Tree models, further fine-tuning of the Logistic Regression model for convergence might not be the most efficient use of resources at this stage.

2. Untuned Decision Tree Performance:

- The untuned Decision Tree exhibited clear signs of overfitting, with a significant drop in performance from the training set (F1-score 99%) to the

test set (F1-score 79.09%). This highlights the necessity of regularization techniques for tree-based models.

3. Pruned Decision Tree Models:

- Both pre-pruning and post-pruning strategies effectively mitigated overfitting and significantly improved the Decision Tree's generalization capabilities.
- The **Pre-pruned Decision Tree** achieved an F1-score of 80% for the 'Canceled' class, an improvement over the untuned version.
- The **Post-pruned Decision Tree** further improved performance, reaching an **F1-score of 81%** for the 'Canceled' class, and also showed the highest overall accuracy (88.06%) and ROC AUC score (0.9283).

VIII.6. Final Model Selection:

Based on the comprehensive comparison, the **Post-pruned Decision Tree Classifier** is selected as the best model for predicting booking cancellations.

Rationale:

- **Highest F1-Score for 'Canceled' Class (81%):** This is crucial for INN Hotels as it provides the best balance between identifying actual cancellations (Recall) and minimizing false alarms (Precision). A higher F1-score directly translates to more effective mitigation of revenue loss and more efficient resource allocation.
- **Superior Overall Accuracy (88.06%) and ROC AUC (0.9283):** These metrics indicate that the model has excellent predictive power and discriminatory ability on unseen data.
- **Robust Balance of Precision (84%) and Recall (78%):** While Recall is vital, a high Precision ensures that the hotel's interventions are targeted, reducing wasted effort on bookings that would not have canceled. The Post-pruned Decision Tree achieves an excellent balance, meaning the hotel can trust its predictions to be mostly accurate while still catching a large proportion of actual cancellations.
- **Mitigation of Overfitting:** The post-pruning technique successfully addressed the overfitting issues observed in the untuned model, resulting in a more reliable and generalizable model.

This Post-pruned Decision Tree Classifier provides a powerful and reliable tool for INN Hotels to proactively manage cancellations, inform refund policies, and ultimately enhance revenue management strategies.

IX. Actionable Recommendations and Business Insights for INN Hotels

Based on the comprehensive Exploratory Data Analysis (EDA) and the robust performance of the Post-pruned Decision Tree Classifier, actionable recommendations and business inferences for INN Hotels are to mitigate revenue loss from cancellations and optimize booking strategies:

1. Leverage the Predictive Model for Proactive Intervention:

- **Implement the Post-pruned Decision Tree Classifier:** Integrate the selected model into the booking system to identify high-risk cancellations in advance. With an 81% F1-score and 78% Recall for cancellations, the model can effectively flag bookings likely to cancel.
- **Tiered Intervention Strategy:**
 - For high-probability cancellations: Initiate early, personalized outreach (e.g., email/SMS reminders, re-confirmation calls, small incentives like a complimentary drink or early check-in). This can re-engage guests and reinforce their commitment.
 - For moderate-probability cancellations: Offer flexible re-booking options or emphasize unique hotel amenities to strengthen their reservation.
- **Dynamic Room Inventory Management:** Use cancellation predictions to dynamically adjust available room inventory. If a room is highly likely to be canceled, the system can cautiously release it back into the market closer to the arrival date, rather than losing potential revenue from an empty room.

2. Optimize Lead Time Management:

- **Analyze Long Lead Time Bookings:** Bookings made with longer lead times show a higher propensity for cancellation. Investigate segments making these bookings and consider implementing specific cancellation policies for them (e.g., requiring a non-refundable deposit after a certain lead time, or offering slight discounts for shorter lead time, non-cancellable bookings).
- **Early Engagement Campaigns:** For bookings with long lead times, implement targeted campaigns that maintain guest excitement and commitment. Share local attraction information, hotel activity schedules, or special offers in the months leading up to their stay.

3. Enhance Customer Loyalty & Retention:

- **Focus on Repeat Guests:** The **1.72% cancellation rate among repeat guests** is exceptionally low, highlighting their high reliability. Invest more in loyalty programs, exclusive offers, and personalized services for this segment to ensure their continued patronage. A very low percentage of current repeated guests (2.5%) indicates significant room for growth in this area.
- **Incentivize First-Time Guests to Become Repeaters:** Implement post-stay surveys and targeted offers for first-time guests to encourage their next booking. A strong first impression is key to building loyalty.

4. Capitalize on Special Requests:

- **Prioritize Special Request Fulfillment:** Guests making special requests are significantly less likely to cancel (mean special requests: 0.76 for Not_Canceled vs. 0.33 for Canceled). Ensure these requests are meticulously fulfilled. This positive experience reinforces commitment and can differentiate INN Hotels.
- **Prompt Communication on Special Requests:** Acknowledge special requests promptly and communicate feasibility. This builds trust and reduces uncertainty, strengthening the booking.

5. Strategic Pricing and Market Segment Focus:

- **Refine Online Channel Strategy:** The 'Online' market segment generates the highest volume and average prices. Continuously optimize online presence, collaborate with OTAs effectively, and ensure competitive pricing. Monitor cancellation rates within this segment closely, as its large volume could heavily impact overall cancellations.
- **Evaluate 'Complementary' Bookings:** The 'Complementary' segment shows very low average prices (often zero). While these might be promotional or internal, their cancellation behavior should be monitored to ensure they align with strategic objectives.
- **Dynamic Pricing Review:** Regularly analyze the relationship between avg_price_per_room and cancellation rates within different market segments. Adjust pricing strategies to find the optimal balance between occupancy, revenue, and cancellation risk.

6. Optimize Seasonal Operations and Staffing:

- **Prepare for Peak Seasons:** October, September, and August are the busiest months. Ensure adequate staffing, inventory, and operational efficiency during these periods to handle increased demand and maintain service quality, which can indirectly reduce cancellations due to dissatisfaction.

- **Targeted Promotions for Off-Peak:** For less busy months, implement targeted marketing and promotional campaigns to attract guests and fill rooms, potentially offering non-refundable deals to secure bookings.

7. Data Collection and Continuous Improvement:

- **Capture Reasons for Cancellation:** Enhance data collection by adding a field for guests to specify reasons for cancellation (if possible). This qualitative data would provide invaluable insights to further refine predictive models and operational strategies.
- **Monitor Model Performance:** Continuously monitor the model's performance metrics (F1-score, Recall, Precision) and retrain it periodically with new data to ensure its accuracy and relevance. The hotel industry is dynamic, and booking patterns can change.

By implementing these recommendations, INN Hotels can move from reactive cancellation management to a proactive, data-driven approach, leading to improved revenue, optimized resource allocation, and enhanced guest satisfaction.